Notes on quantile regression via proximal MCMC

1 Bayesian Formulation for Individual Quantile Trend Filtering

1.1 Model Setup and Likelihood

Consider the observed time series $\mathbf{y} = (y_1, y_2, \ldots, y_n)^T$ and let $\tau \in (0, 1)$ be a single quantile level of interest. We seek to estimate the quantile function $\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_n)^T$, where θ_i represents the τ-th quantile of y_i.

The likelihood for quantile regression is derived from the asymmetric Laplace distribution (ALD), which provides the connection between the check function and a proper probability distribution. The ALD has density:

$$f(y_i|\theta_i, \sigma, \tau) = \frac{\tau(1-\tau)}{\sigma} \exp\left(-\frac{\rho_\tau(y_i - \theta_i)}{\sigma}\right)$$

The joint likelihood for all observations is:

$$\mathcal{L}(\boldsymbol{\theta}|\mathbf{y}) = \prod_{i=1}^{n} f(y_i|\theta_i, \tau) = \prod_{i=1}^{n} \frac{\tau(1-\tau)}{\sigma} \exp\left(-\frac{\rho_\tau(y_i - \theta_i)}{\sigma}\right)$$

1.2 Prior Specifications

1.2.1 Prior on Quantile Function

We impose smoothness on the quantile function through a trend filtering prior. Let $\mathbf{D}^{(k+1)} \in \mathbb{R}^{(n-k-1)\times n}$ be the $(k+1)$-th order discrete difference operator. The prior on $\boldsymbol{\theta}$ is:

$$\pi(\boldsymbol{\theta}|\lambda) \propto \exp\left(-\lambda \|\mathbf{D}^{(k+1)}\boldsymbol{\theta}\|_1\right)$$

This corresponds to independent Laplace priors on the $(k+1)$-th order differences:

$$(\mathbf{D}^{(k+1)}\boldsymbol{\theta})_\ell \overset{\text{iid}}{\sim} \text{Laplace}\left(0, \frac{1}{\lambda}\right), \quad \ell = 1, \ldots, n-k-1$$

1.2.2 Prior on Regularization Parameter

We place a Gamma prior on the regularization parameter λ:

$$\lambda \sim \text{Gamma}(a, b)$$

with density:

$$\pi(\lambda) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda}, \quad \lambda > 0$$

where $a > 0$ is the shape parameter and $b > 0$ is the rate parameter.

1.3 Posterior Distribution

Applying Bayes' theorem, the joint posterior distribution is:

$$\begin{aligned}\pi(\boldsymbol{\theta}, \lambda|\mathbf{y}) &\propto \mathcal{L}(\boldsymbol{\theta}|\mathbf{y})\pi(\boldsymbol{\theta}|\lambda)\pi(\lambda) \\ &\propto \exp\left(-\sum_{i=1}^{n} \frac{\rho_\tau(y_i - \theta_i)}{\sigma}\right) \exp\left(-\lambda\|\mathbf{D}^{(k+1)}\boldsymbol{\theta}\|_1\right) \lambda^{a-1}e^{-b\lambda}\end{aligned}$$

Taking the negative logarithm (up to constants), the negative log-posterior is:

$$-\log \pi(\boldsymbol{\theta}, \lambda|\mathbf{y}) = \sum_{i=1}^{n} \frac{\rho_\tau(y_i - \theta_i)}{\sigma} + \lambda\|\mathbf{D}^{(k+1)}\boldsymbol{\theta}\|_1 - (a-1)\log\lambda + b\lambda + C$$

where C is a normalization constant.

1.4 Conditional Posterior Distributions

1.4.1 Conditional Posterior of θ given λ

The conditional posterior of $\boldsymbol{\theta}$ given λ is:

$$\pi(\boldsymbol{\theta}|\lambda, \mathbf{y}) \propto \exp\left(-\sum_{i=1}^{n} \frac{\rho_\tau(y_i - \theta_i)}{\sigma} - \lambda\|\mathbf{D}^{(k+1)}\boldsymbol{\theta}\|_1\right)$$

This distribution is non-standard due to the non-smooth ℓ_1 penalty, necessitating specialized MCMC methods such as P-MALA (Proximal Metropolis-Adjusted Langevin Algorithm).

1.4.2 Conditional Posterior of λ given θ

The conditional posterior of λ given $\boldsymbol{\theta}$ is derived as follows. From the joint posterior, the terms involving λ are:

$$\pi(\lambda|\boldsymbol{\theta}, \mathbf{y}) \propto \exp(-\lambda S) \cdot \lambda^{a-1}e^{-b\lambda}$$

where $S = \|\mathbf{D}^{(k+1)}\boldsymbol{\theta}\|_1$ is the total variation penalty.
Combining the exponential terms:

$$\pi(\lambda|\boldsymbol{\theta}, \mathbf{y}) \propto \lambda^{a-1}\exp(-(b+S)\lambda)$$

This is the kernel of a Gamma distribution. Therefore:

$$\lambda|\boldsymbol{\theta}, \mathbf{y} \sim \text{Gamma}(a, b+S)$$

The posterior shape parameter remains a, while the rate parameter becomes $b + S$, where S represents the total absolute variation in the $(k+1)$-th order differences of the current quantile estimate.

1.5 Gibbs Sampling Algorithm

The posterior sampling proceeds via a two-step Gibbs sampler:
The Gibbs sampler for individual quantile trend filtering proceeds as follows:

1. Initialize $\boldsymbol{\theta}^{(0)}$ and $\lambda^{(0)}$

2. For $t = 1, 2, \ldots, T$:

   - Sample $\boldsymbol{\theta}^{(t)}|\lambda^{(t-1)}, \mathbf{y}$ using P-MALA
   - Sample $\lambda^{(t)}|\boldsymbol{\theta}^{(t)}, \mathbf{y} \sim \text{Gamma}(a, b + S^{(t)})$ where $S^{(t)} = \|\mathbf{D}^{(k+1)}\boldsymbol{\theta}^{(t)}\|_1$

The P-MALA step for $\boldsymbol{\theta}$ involves: 1. Langevin proposal using the gradient of a smooth approximation to the check function 2. Application of the proximal operator for the trend filtering penalty 3. Metropolis-Hastings acceptance/rejection step

The Gamma sampling step for λ is exact and can be performed using standard random number generators.

1.6 Conjugacy and Computational Advantages

The key computational advantage of this formulation is the conjugacy between the Gamma prior on λ and the Laplace prior on the differences $\mathbf{D}^{(k+1)}\boldsymbol{\theta}$. This conjugacy arises because:

- The Laplace distribution on differences contributes $\exp(-\lambda S)$ to the likelihood
- The Gamma prior contributes $\lambda^{a-1}e^{-b\lambda}$
- Their product yields another Gamma distribution with updated rate parameter $b + S$

This allows for exact sampling of λ in each Gibbs iteration, while the more complex sampling of $\boldsymbol{\theta}$ is handled by the sophisticated P-MALA algorithm that can deal with the non-smooth penalty and potentially large dimensionality.

This framework provides a fully Bayesian approach to individual quantile trend filtering that properly accounts for uncertainty in both the quantile function and the regularization parameter.

2 Computational Efficiency Through Smart Initialization

2.1 Data-Dependent Prior Specification

Rather than using fixed hyperparameters, we set the Gamma prior on the penalty parameter λ to scale with sample size:

$$\begin{gathered}\lambda \sim \text{Gamma}(a, b) \\ \text{where } a = \left(\frac{\mu_\lambda}{\sigma_\lambda}\right)^2, \quad b = \frac{\mu_\lambda}{\sigma_\lambda^2} \\ \mu_\lambda = 3n, \quad \sigma_\lambda = n\end{gathered}$$

This choice is motivated by:

2.2 Empirical Validation

To validate our approach across diverse signal characteristics, we tested three baseline functions designed to represent scenarios commonly encountered in real data applications:

1. **Smooth monotonic trend:** $f(x) = \sqrt{x/n}$, representing gradually changing baselines with minimal curvature
2. **Moderate frequency oscillation:** $f(x) = \sin(2\pi x/n)$, representing seasonal or cyclical patterns
3. **High frequency oscillation:** $f(x) = \sin(10\pi x/n) + \cos(15\pi x/n)$, representing complex multi-scale variations

For each baseline function, we generated noisy observations:

$$y_i = f(x_i) + \epsilon_i, \quad i = 1, \ldots, 1000$$
$$\epsilon_i \sim N(0, \sigma^2)$$

where $\sigma = 0.2$ for the smooth monotonic trend and $\sigma = 0.4$ for both oscillatory functions. These functions were deliberately chosen to span the spectrum from smooth to highly variable, allowing us to assess the robustness of our initialization strategy and data-dependent priors across substantially different signal characteristics that arise in practice.

2.3 Key Findings

We validated our approach by running 20 independent replications for each of the three baseline functions, with different random seeds generating distinct noise realizations. For each replication, we ran 8,000 Gibbs iterations with 3,000 burn-in, using data-dependent priors $\mu_\lambda = 3n$ and $\sigma_\lambda = n$. All tests used $n = 1000$ observations and estimated the median quantile ($\tau = 0.5$).

2.3.1 Adaptive Penalty Selection

The most striking result is the method's adaptivity across substantially different function types. Table 1 shows the posterior mean of λ averaged across 20 datasets for each function:

Function	**Mean**	**Median**	**Mode**
Smooth monotonic ($\sqrt{x/n}$)	1579	1501	1343
Moderate frequency ($\sin(2\pi x/n)$)	1639	1561	1409
High frequency ($\sin(10\pi x/n) + \cos(15\pi x/n)$)	124	118	108

Table 1: Posterior λ statistics averaged across 20 independent datasets. The method automatically selects substantially larger penalty parameters for smoother functions.

The algorithm demonstrates strong adaptivity: smooth and moderate-frequency functions receive penalty parameters approximately **13 times larger** than the high-frequency function. This automatic calibration reflects the fundamental principle of trend filtering—smoother underlying functions require stronger regularization to prevent overfitting to noise, while highly variable functions need lighter regularization to capture genuine signal variation.

2.3.2 Estimation Accuracy

Figure (1) shows the averaged θ estimates across 20 datasets for each function. The estimates (red dashed) closely track the theoretical quantile curves (black solid) across all function types, demonstrating the robustness of our initialization strategy and data-dependent priors.

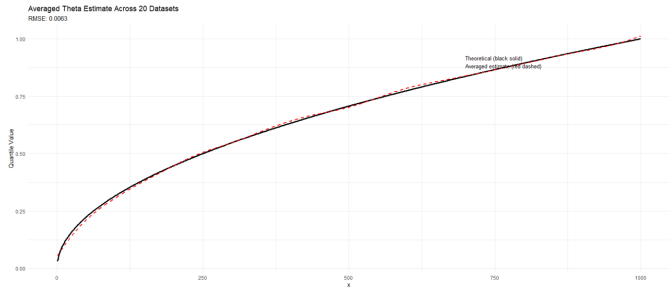
Table (1) summarizes the root mean squared error (RMSE) for each function type:

The smooth monotonic function achieves the lowest RMSE (0.0063), benefiting from both its inherent smoothness and lower noise level ($\sigma = 0.2$ versus $\sigma = 0.4$ for oscillatory functions). The high-frequency function shows slightly higher RMSE (0.0508) but still provides excellent recovery of the complex multi-scale variation.

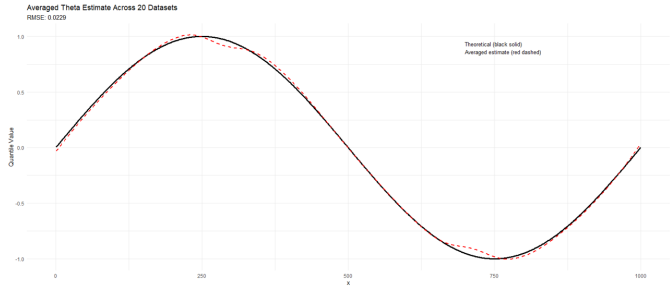
2.3.3 Stability Across Replications

Examining individual dataset results reveals consistent performance:

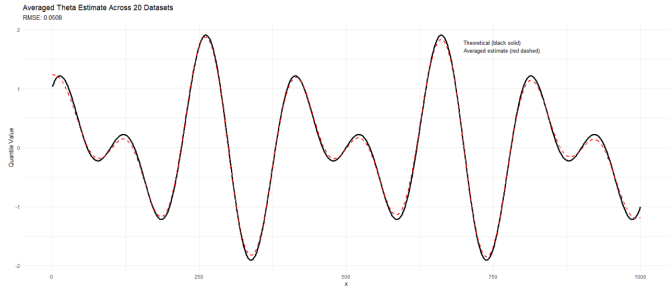
- **Smooth monotonic**: Individual RMSEs ranged from 0.0142 to 0.0353, with λ posterior means ranging from 1530 to 1630. Despite the 10-fold variation in RMSE, λ estimates remained stable within $\pm 6\%$ of the mean.
- **High frequency**: Individual RMSEs ranged from 0.0692 to 0.1127, with λ posterior means ranging from 110 to 132. The coefficient of variation in λ was approximately 6%, indicating robust parameter estimation across noise realizations.



(a) Smooth monotonic



(b) Moderate frequency



(c) High frequency

Figure 1: Averaged theta estimates (red dashed) versus theoretical quantile curves (black solid) across 20 independent datasets.

This stability demonstrates that the data-dependent prior specification ($\mu_\lambda = 3n$, $\sigma_\lambda = n$) successfully scales across different sample sizes and function characteristics without requiring manual tuning.

2.3.4 Computational Efficiency

The optimized configuration achieves these results with:

- 8,000 total Gibbs iterations (3,000 burn-in)
- Single P-MALA sample per Gibbs iteration (no inner burn-in)
- Acceptance rates consistently in the range 0.45–0.62 across all function types
- Initialization using $\theta^{(0)} = y$ (raw data)

2.3.5 Summary

The empirical validation establishes four key properties of our approach:

1. **Adaptivity**: Automatic penalty selection appropriately scales to function roughness, with λ varying by over an order of magnitude between smooth and highly oscillatory functions.

2. **Accuracy**: RMSE values confirm excellent quantile recovery across diverse function types, from smooth monotonic trends to complex multi-frequency oscillations.

3. **Stability**: Consistent λ estimates across independent replications demonstrate robustness to random noise realizations.

These findings validate the practical utility of combining Proximal MCMC with conjugate Gamma priors and preprocessing-based initialization for Bayesian quantile trend filtering.

3 Bayesian Quantile Trend Filtering via Hierarchical Priors

3.1 Hierarchical Structure

We consider the following hierarchical Bayesian model for quantile trend filtering:

$$\begin{aligned} y|\theta &\sim \text{Quantile Likelihood with parameter } \tau \\ (D\theta)_j|\gamma_j^2 &\sim N(0,\gamma_j^2), \quad j=1,\ldots,m \\ \gamma_j^2|\lambda &\overset{\text{i.i.d.}}{\sim} \text{Exp}(\lambda^2/2), \quad j=1,\ldots,m \\ \lambda^2 &\sim \text{Gamma}(a,b) \end{aligned}$$

where:

- $y=(y_1,\ldots,y_n)^T$ is the vector of observations
- $\theta=(\theta_1,\ldots,\theta_n)^T$ is the vector of trend parameters
- $D\in\mathbb{R}^{m\times n}$ is the $(k+1)$-th order difference matrix with $m=n-k-1$
- $\tau\in(0,1)$ is the quantile level of interest
- $\{\gamma_j^2\}_{j=1}^m$ are variance parameters for each difference
- $\lambda>0$ is the penalty parameter
- $a,b>0$ are hyperparameters

3.2 Distributional Forms

3.2.1 Quantile Likelihood

The likelihood based on the check function is:

$$p(y|\theta)\propto\exp\left(-\sum_{i=1}^{n}\rho_\tau(y_i-\theta_i)\right)$$

where $\rho_\tau(u)=u[\tau-\mathbf{1}(u<0)]$ is the check function.

3.2.2 Normal Prior on Differences

For each $j=1,\ldots,m$:

$$p((D\theta)_j|\gamma_j^2)=\frac{1}{\sqrt{2\pi\gamma_j^2}}\exp\left(-\frac{(D\theta)_j^2}{2\gamma_j^2}\right)$$

The joint prior on θ given $\{\gamma_j^2\}$ is:

$$p(\theta|\{\gamma_j^2\})=\prod_{j=1}^{m}p((D\theta)_j|\gamma_j^2)=\prod_{j=1}^{m}\frac{1}{\sqrt{2\pi\gamma_j^2}}\exp\left(-\frac{(D\theta)_j^2}{2\gamma_j^2}\right)$$

3.2.3 Exponential Prior on Variances

For each $j = 1, \ldots, m$:

$$p(\gamma_j^2|\lambda) = \frac{\lambda^2}{2} \exp\left(-\frac{\lambda^2\gamma_j^2}{2}\right), \quad \gamma_j^2 > 0$$

The joint prior on $\{\gamma_j^2\}$ given λ is:

$$p(\{\gamma_j^2\}|\lambda) = \prod_{j=1}^{m} \frac{\lambda^2}{2} \exp\left(-\frac{\lambda^2\gamma_j^2}{2}\right) = \left(\frac{\lambda^2}{2}\right)^m \exp\left(-\frac{\lambda^2}{2}\sum_{j=1}^{m}\gamma_j^2\right)$$

3.2.4 Gamma Prior on Penalty Parameter

$$p(\lambda^2) = \frac{b^a}{\Gamma(a)}(\lambda^2)^{a-1}\exp(-b\lambda^2), \quad \lambda^2 > 0$$

3.3 Joint Distribution

The joint distribution of all variables is:

$$\begin{aligned} p(y, \theta, \{\gamma_j^2\}, \lambda^2) &= p(y|\theta) \cdot p(\theta|\{\gamma_j^2\}) \cdot p(\{\gamma_j^2\}|\lambda) \cdot p(\lambda^2) \\ &\propto \exp\left(-\sum_{i=1}^{n}\rho_\tau(y_i - \theta_i)\right) \cdot \prod_{j=1}^{m} \frac{1}{\sqrt{2\pi\gamma_j^2}} \exp\left(-\frac{(D\theta)_j^2}{2\gamma_j^2}\right) \\ &\quad \times \left(\frac{\lambda^2}{2}\right)^m \exp\left(-\frac{\lambda^2}{2}\sum_{j=1}^{m}\gamma_j^2\right) \cdot \frac{b^a}{\Gamma(a)}(\lambda^2)^{a-1}\exp(-b\lambda^2) \end{aligned}$$

4 Derivation of Full Conditional Distributions

4.1 Full Conditional: $p(\theta|y, \{\gamma_j^2\}, \lambda^2)$

4.1.1 Step 1: Apply Bayes' Theorem

By Bayes' theorem, the full conditional is:

$$p(\theta|y, \{\gamma_j^2\}, \lambda^2) = \frac{p(y, \theta, \{\gamma_j^2\}, \lambda^2)}{p(y, \{\gamma_j^2\}, \lambda^2)}$$

4.1.2 Step 2: Identify Relevant Terms

The denominator does not depend on θ, so:

$$p(\theta|y, \{\gamma_j^2\}, \lambda^2) \propto p(y, \theta, \{\gamma_j^2\}, \lambda^2)$$

From the joint distribution, only the terms involving θ are relevant:

$$p(\theta|y, \{\gamma_j^2\}, \lambda^2) \propto p(y|\theta) \cdot p(\theta|\{\gamma_j^2\})$$

The terms $p(\{\gamma_j^2\}|\lambda)$ and $p(\lambda^2)$ do not depend on θ and are absorbed into the normalizing constant.

4.1.3 Step 3: Write the Likelihood

$$p(y|\theta) \propto \exp\left(-\sum_{i=1}^{n}\rho_\tau(y_i - \theta_i)\right)$$

4.1.4 Step 4: Write the Prior on θ

$$p(\theta|\{\gamma_j^2\}) = \prod_{j=1}^{m} \frac{1}{\sqrt{2\pi\gamma_j^2}} \exp\left(-\frac{(D\theta)_j^2}{2\gamma_j^2}\right) \propto \exp\left(-\sum_{j=1}^{m} \frac{(D\theta)_j^2}{2\gamma_j^2}\right)$$

4.1.5 Step 5: Combine Terms

$$\begin{aligned} p(\theta|y, \{\gamma_j^2\}, \lambda^2) &\propto \exp\left(-\sum_{i=1}^{n} \rho_\tau(y_i - \theta_i)\right) \cdot \exp\left(-\sum_{j=1}^{m} \frac{(D\theta)_j^2}{2\gamma_j^2}\right) \\ &= \exp\left(-\sum_{i=1}^{n} \rho_\tau(y_i - \theta_i) - \sum_{j=1}^{m} \frac{(D\theta)_j^2}{2\gamma_j^2}\right) \end{aligned}$$

4.1.6 Result

$$\boxed{p(\theta|y, \{\gamma_j^2\}, \lambda^2) \propto \exp\left(-\sum_{i=1}^{n} \rho_\tau(y_i - \theta_i) - \sum_{j=1}^{m} \frac{(D\theta)_j^2}{2\gamma_j^2}\right)}$$

The negative log-posterior has the form $f(\theta) + g(\theta)$ where:

- $f(\theta) = \sum_{i=1}^{n} \rho_\tau(y_i - \theta_i)$ is the data fidelity term
- $g(\theta) = \sum_{j=1}^{m} \frac{(D\theta)_j^2}{2\gamma_j^2} = \frac{1}{2}(D\theta)^T D_\gamma^{-1}(D\theta)$ is the penalty term

where $D_\gamma = \text{diag}(\gamma_1^2, \ldots, \gamma_m^2)$.

This is not a standard distribution and requires Proximal MCMC (P-MALA) for sampling.

4.2 Full Conditional: $p(\gamma_j^2|y, \theta, \{\gamma_{k\neq j}^2\}, \lambda^2)$

4.2.1 Step 1: Apply Bayes' Theorem

The full conditional for γ_j^2 is:

$$p(\gamma_j^2|y, \theta, \{\gamma_{k\neq j}^2\}, \lambda^2) \propto p(y, \theta, \{\gamma_j^2\}, \lambda^2)$$

4.2.2 Step 2: Identify Relevant Terms

Only terms involving γ_j^2 are relevant. From the joint distribution:

- $p(y|\theta)$ does not depend on γ_j^2 (drops out)
- $p(\theta|\{\gamma_j^2\})$ contains $p((D\theta)_j|\gamma_j^2)$
- $p(\{\gamma_j^2\}|\lambda)$ contains $p(\gamma_j^2|\lambda)$
- $p(\lambda^2)$ does not depend on γ_j^2 (drops out)

Therefore:

$$p(\gamma_j^2|y, \theta, \{\gamma_{k\neq j}^2\}, \lambda^2) \propto p((D\theta)_j|\gamma_j^2) \cdot p(\gamma_j^2|\lambda)$$

4.2.3 Step 3: Write $p((D\theta)_j|\gamma_j^2)$

$$p((D\theta)_j|\gamma_j^2) = \frac{1}{\sqrt{2\pi\gamma_j^2}} \exp\left(-\frac{(D\theta)_j^2}{2\gamma_j^2}\right) \propto (\gamma_j^2)^{-1/2} \exp\left(-\frac{(D\theta)_j^2}{2\gamma_j^2}\right)$$

4.2.4 Step 4: Write $p(\gamma_j^2|\lambda)$

$$p(\gamma_j^2|\lambda) = \frac{\lambda^2}{2}\exp\left(-\frac{\lambda^2\gamma_j^2}{2}\right) \propto \exp\left(-\frac{\lambda^2\gamma_j^2}{2}\right)$$

4.2.5 Step 5: Combine Terms

$$\begin{aligned} p(\gamma_j^2|y,\theta,\{\gamma_{k\neq j}^2\},\lambda^2) &\propto (\gamma_j^2)^{-1/2}\exp\left(-\frac{(D\theta)_j^2}{2\gamma_j^2}\right)\cdot\exp\left(-\frac{\lambda^2\gamma_j^2}{2}\right) \\ &= (\gamma_j^2)^{-1/2}\exp\left(-\frac{(D\theta)_j^2}{2\gamma_j^2}-\frac{\lambda^2\gamma_j^2}{2}\right) \end{aligned}$$

4.2.6 Step 6: Recognize as Generalized Inverse-Gaussian

This has the form:

$$x^{-1/2}\exp\left(-\frac{a}{2x}-\frac{bx}{2}\right)$$

with $x=\gamma_j^2$, $a=(D\theta)_j^2$, and $b=\lambda^2$.

The Generalized Inverse-Gaussian distribution $\text{GIG}(p,a,b)$ has the kernel:

$$f(x;p,a,b) \propto x^{p-1}\exp\left(-\frac{1}{2}\left(\frac{a}{x}+bx\right)\right)$$

For $p=1/2$:

$$f(x;1/2,a,b) \propto x^{1/2-1}\exp\left(-\frac{1}{2}\left(\frac{a}{x}+bx\right)\right) = x^{-1/2}\exp\left(-\frac{a}{2x}-\frac{bx}{2}\right)$$

4.2.7 Result

$$\boxed{\gamma_j^2|y,\theta,\{\gamma_{k\neq j}^2\},\lambda^2 \sim \text{GIG}\left(p=\frac{1}{2}, a=(D\theta)_j^2, b=\lambda^2\right)}$$

where GIG denotes the Generalized Inverse-Gaussian distribution with parameters (p,a,b). The γ_j^2 are conditionally independent given θ and λ^2.

4.3 Full Conditional: $p(\lambda^2|y,\theta,\{\gamma_j^2\})$

4.3.1 Step 1: Apply Bayes' Theorem

The full conditional for λ^2 is:

$$p(\lambda^2|y,\theta,\{\gamma_j^2\}) \propto p(y,\theta,\{\gamma_j^2\},\lambda^2)$$

4.3.2 Step 2: Identify Relevant Terms

Only terms involving λ^2 are relevant:

- $p(y|\theta)$ does not depend on λ^2 (drops out)
- $p(\theta|\{\gamma_j^2\})$ does not depend on λ^2 (drops out)
- $p(\{\gamma_j^2\}|\lambda)$ depends on λ^2
- $p(\lambda^2)$ is the prior on λ^2

Therefore:

$$p(\lambda^2|y,\theta,\{\gamma_j^2\}) \propto p(\{\gamma_j^2\}|\lambda)\cdot p(\lambda^2)$$

4.3.3 Step 3: Write $p(\{\gamma_j^2\}|\lambda)$

$$\begin{aligned} p(\{\gamma_j^2\}|\lambda) &= \prod_{j=1}^{m} \frac{\lambda^2}{2} \exp\left(-\frac{\lambda^2 \gamma_j^2}{2}\right) \\ &= \left(\frac{\lambda^2}{2}\right)^m \exp\left(-\frac{\lambda^2}{2} \sum_{j=1}^{m} \gamma_j^2\right) \\ &\propto (\lambda^2)^m \exp\left(-\frac{\lambda^2}{2} \sum_{j=1}^{m} \gamma_j^2\right) \end{aligned}$$

4.3.4 Step 4: Write $p(\lambda^2)$

$$p(\lambda^2) = \frac{b^a}{\Gamma(a)} (\lambda^2)^{a-1} \exp(-b\lambda^2) \propto (\lambda^2)^{a-1} \exp(-b\lambda^2)$$

4.3.5 Step 5: Combine Terms

$$\begin{aligned} p(\lambda^2|y, \theta, \{\gamma_j^2\}) &\propto (\lambda^2)^m \exp\left(-\frac{\lambda^2}{2} \sum_{j=1}^{m} \gamma_j^2\right) \cdot (\lambda^2)^{a-1} \exp(-b\lambda^2) \\ &= (\lambda^2)^m \cdot (\lambda^2)^{a-1} \cdot \exp\left(-\frac{\lambda^2}{2} \sum_{j=1}^{m} \gamma_j^2 - b\lambda^2\right) \\ &= (\lambda^2)^{m+a-1} \cdot \exp\left(-\lambda^2 \left(\frac{1}{2} \sum_{j=1}^{m} \gamma_j^2 + b\right)\right) \end{aligned}$$

4.3.6 Step 6: Recognize as Gamma

This has the form:

$$x^{\alpha-1} \exp(-\beta x)$$

which is the kernel of a Gamma distribution Gamma(α, β) with shape α and rate β.

In our case:

$$\begin{aligned} \alpha &= m + a \\ \beta &= b + \frac{1}{2} \sum_{j=1}^{m} \gamma_j^2 \end{aligned}$$

4.3.7 Result

$$\boxed{\lambda^2|y, \theta, \{\gamma_j^2\} \sim \text{Gamma}\left(a + m,\ b + \frac{1}{2} \sum_{j=1}^{m} \gamma_j^2\right)}$$

4.4 Prior on λ^2

In the hierarchical formulation, we place a Gamma prior on λ^2 (not on λ):

$$\lambda^2 \sim \text{Gamma}(a, b)$$

with density

$$p(\lambda^2) = \frac{b^a}{\Gamma(a)} (\lambda^2)^{a-1} \exp(-b\lambda^2), \quad \lambda^2 > 0.$$

Why not place the prior on λ directly?

At first glance, it might seem more natural to place a Gamma prior directly on λ rather than λ^2. However, this choice breaks the conjugacy structure that makes Gibbs sampling tractable.

To see why, consider the full conditional for the penalty parameter. The relevant terms from the joint distribution are:

$$p(\lambda|\theta, \{\gamma_j^2\}) \propto p(\{\gamma_j^2\}|\lambda) \cdot p(\lambda).$$

Recall that the prior on the variance parameters is

$$p(\{\gamma_j^2\}|\lambda) = \prod_{j=1}^{m} \frac{\lambda^2}{2} \exp\left(-\frac{\lambda^2\gamma_j^2}{2}\right) = \left(\frac{\lambda^2}{2}\right)^m \exp\left(-\frac{\lambda^2}{2}\sum_{j=1}^{m}\gamma_j^2\right).$$

Case 1: Prior on λ^2 (conjugate)

If we place a Gamma prior on λ^2:

$$p(\lambda^2) \propto (\lambda^2)^{a-1} \exp(-b\lambda^2),$$

then the full conditional becomes:

$$\begin{aligned} p(\lambda^2|\theta, \{\gamma_j^2\}) &\propto (\lambda^2)^m \exp\left(-\frac{\lambda^2}{2}\sum_{j=1}^{m}\gamma_j^2\right) \cdot (\lambda^2)^{a-1} \exp(-b\lambda^2) \\ &= (\lambda^2)^{m+a-1} \exp\left(-\lambda^2\left(b + \frac{1}{2}\sum_{j=1}^{m}\gamma_j^2\right)\right). \end{aligned}$$

This is the kernel of a Gamma distribution:

$$\lambda^2|\theta, \{\gamma_j^2\} \sim \text{Gamma}\left(a+m,\, b + \frac{1}{2}\sum_{j=1}^{m}\gamma_j^2\right).$$

We can sample this exactly using standard algorithms.

Case 2: Prior on λ (non-conjugate)

If instead we place a Gamma prior on λ:

$$p(\lambda) \propto \lambda^{a-1} \exp(-b\lambda),$$

then the full conditional becomes:

$$\begin{aligned} p(\lambda|\theta, \{\gamma_j^2\}) &\propto (\lambda^2)^m \exp\left(-\frac{\lambda^2}{2}\sum_{j=1}^{m}\gamma_j^2\right) \cdot \lambda^{a-1} \exp(-b\lambda) \\ &= \lambda^{2m} \exp\left(-\frac{\lambda^2}{2}\sum_{j=1}^{m}\gamma_j^2\right) \cdot \lambda^{a-1} \exp(-b\lambda) \\ &= \lambda^{2m+a-1} \exp\left(-\frac{\lambda^2}{2}\sum_{j=1}^{m}\gamma_j^2 - b\lambda\right). \end{aligned}$$

This distribution does *not* belong to any standard family due to the mixed λ and λ^2 terms in the exponent. The full conditional is non-conjugate, and we would need Metropolis-Hastings or another MCMC method to sample λ, losing the computational advantage of exact Gibbs sampling.

Conclusion: The prior must be placed on λ^2 to maintain conjugacy with the exponential prior on the variance parameters $\{\gamma_j^2\}$. This conjugacy is essential for efficient Gibbs sampling, allowing us to sample λ^2 exactly from a Gamma distribution at each iteration.

5 Summary of Full Conditionals

For Gibbs sampling, we have the following full conditional distributions:

1. **Trend parameter:**

$$p(\theta|y, \{\gamma_j^2\}, \lambda^2) \propto \exp\left(-\sum_{i=1}^{n} \rho_\tau(y_i - \theta_i) - \sum_{j=1}^{m} \frac{(D\theta)_j^2}{2\gamma_j^2}\right)$$

   Sample using Proximal MCMC (P-MALA).

2. **Variance parameters (conditionally independent):**

$$\gamma_j^2|y, \theta, \lambda^2 \sim \text{GIG}\left(\frac{1}{2}, (D\theta)_j^2, \lambda^2\right), \quad j = 1, \ldots, m$$

3. **Penalty parameter:**

$$\lambda^2|y, \theta, \{\gamma_j^2\} \sim \text{Gamma}\left(a + m,\, b + \frac{1}{2}\sum_{j=1}^{m} \gamma_j^2\right)$$

6 Connection to Lasso Penalty

By marginalizing out the variance parameters $\{\gamma_j^2\}$, the conditional prior on $(D\theta)_j$ becomes:

$$p((D\theta)_j|\lambda) = \int_0^\infty p((D\theta)_j|\gamma_j^2)p(\gamma_j^2|\lambda)d\gamma_j^2 = \frac{\lambda}{2}\exp(-\lambda|(D\theta)_j|)$$

This is the Laplace (double exponential) distribution, which yields the L1 penalty:

$$p(D\theta|\lambda) = \prod_{j=1}^{m} \frac{\lambda}{2}\exp(-\lambda|(D\theta)_j|) \propto \exp(-\lambda\|D\theta\|_1)$$

Therefore, marginalizing over $\{\gamma_j^2\}$ and λ^2 gives:

$$p(\theta|y) \propto \exp\left(-\sum_{i=1}^{n} \rho_\tau(y_i - \theta_i)\right) \cdot h(\|D\theta\|_1)$$

where h is a function that penalizes the L1 norm of the differences, implementing the Lasso-type penalty in the Bayesian framework.

7 Why the Hierarchical Approach Fails for Quantile Trend Filtering

7.1 Motivation

The hierarchical formulation presented in Section 3, inspired by Park & Casella's Bayesian Lasso, offers theoretical elegance through its scale mixture representation. By placing Normal priors on differences conditional on variance parameters γ_j^2, and Exponential priors on these variances, the marginalized prior recovers the desired Laplace (L1) penalty. This approach provides local adaptivity in principle, as each difference $(D\theta)_j$ receives its own variance parameter γ_j^2 that can adapt to local function roughness.

However, despite this theoretical appeal and successful application in linear regression contexts, we found this hierarchical approach fundamentally unsuitable for quantile trend filtering. We now explain why.

7.2 Empirical Observation: Universal Collapse

We implemented the hierarchical Gibbs sampler with full conditionals derived in Section 3 and tested it on three baseline functions designed to span the spectrum from smooth to highly variable:

1. **Smooth monotonic:** $f(x) = \sqrt{x/n}$
2. **Moderate frequency:** $f(x) = \sin(2\pi x/n)$
3. **High frequency:** $f(x) = \sin(10\pi x/n) + \cos(15\pi x/n)$

Across all three functions, we observed that the squared second-order differences $(D\theta)_j^2$ collapsed to extremely small values:

$$|(D\theta)_j|^2 \approx 10^{-10} \text{ to } 10^{-14}, \quad \forall j = 1, \ldots, m$$

This occurred regardless of the true underlying function's roughness, and persisted even when initializing with moderate penalty values (e.g., $\lambda = 100$).

7.3 Mechanistic Explanation: Pathological Feedback Loop

The collapse of all variance parameters stems from the properties of the Generalized Inverse Gaussian (GIG) distribution. Recall that the full conditional for each variance parameter is:

$$\gamma_j^2 | y, \theta, \lambda^2 \sim \text{GIG}\left(\frac{1}{2}, (D\theta)_j^2, \lambda^2\right)$$

The GIG distribution has mean and variance:

$$\mathbb{E}[\gamma_j^2] = \sqrt{\frac{(D\theta)_j^2}{\lambda^2}} \cdot \frac{K_{3/2}(\sqrt{\lambda^2 (D\theta)_j^2})}{K_{1/2}(\sqrt{\lambda^2 (D\theta)_j^2})}$$

$$\text{Var}[\gamma_j^2] \propto \frac{(D\theta)_j^2}{\lambda^2}$$

When $(D\theta)_j^2$ is extremely small (on the order of 10^{-12} to 10^{-14}) and λ^2 is moderate to large, both the mean and variance of γ_j^2 collapse toward zero. The variance parameter becomes concentrated around zero regardless of the actual local roughness of the function.

7.4 Loss of Adaptivity: The Fatal Flaw

The fundamental problem is that $(D\theta)_j^2$ **is very small for both smooth and rough functions**. This creates a pathological situation:

- For genuinely smooth functions (e.g., $\sqrt{x/n}$), small $(D\theta)_j^2$ values are appropriate the second order differences should be small.
- For highly oscillatory functions (e.g., $\sin(10\pi x/n) + \cos(15\pi x/n)$), the $(D\theta)_j^2$ values are also small, even though these functions exhibit substantial local variation.

The hierarchical model cannot distinguish between these fundamentally different scenarios. When global smoothing forces θ to be smooth, all differences approach zero, causing all variance parameters γ_j^2 to collapse regardless of the true function's roughness. This creates a pathological feedback loop:

1. Small $(D\theta)_j^2$ forces small γ_j^2 via the GIG conditional
2. Small γ_j^2 enforces strong local smoothing through the penalty term $\sum_j \frac{(D\theta)_j^2}{2\gamma_j^2}$
3. Strong local smoothing produces even smaller $(D\theta)_j^2$ in the next iteration

This feedback loop drives the system toward universal smoothing, making the model non-adaptive. The local variance parameters, which should provide adaptivity by varying across j to reflect local function characteristics, instead all collapse toward zero together.

8 ADMM Solution for the Constrained Proximal Operator

Now we move to joint quantile estimation, in which case non-crossing constrainst between the quantiles must be satisfied, we begin with two different quantiles as a proof of concept.

8.1 Problem Formulation

In the P-MALA algorithm for joint quantile estimation, we must compute the proximal operator at each iteration. For $K = 2$ quantiles with parameters $\theta = (\theta^{(1)}, \theta^{(2)}) \in \mathbb{R}^{2n}$, the prior distribution is:

$$\pi(\theta|\lambda_1, \lambda_2) \propto \exp\left(-\lambda_1\|D\theta^{(1)}\|_1 - \lambda_2\|D\theta^{(2)}\|_1\right) \cdot \mathbb{I}_{\mathcal{C}}(\theta) \tag{1}$$

where $D \in \mathbb{R}^{m\times n}$ is the $(k+1)$-th order difference matrix, $\lambda_1, \lambda_2 > 0$ are penalty parameters, and $\mathcal{C} = \{\theta : \theta_i^{(1)} \le \theta_i^{(2)}, \forall i = 1, \ldots, n\}$ is the non-crossing constraint set.

The proximal operator we must compute is:

$$\text{prox}_{\gamma g}(w) = \arg\min_{\theta\in\mathbb{R}^{2n}} \left\{ \frac{1}{2\gamma}\|\theta - w\|_2^2 + g(\theta) \right\} \tag{2}$$

where $w \in \mathbb{R}^{2n}$ is the Langevin proposal and

$$g(\theta) = \lambda_1\|D\theta^{(1)}\|_1 + \lambda_2\|D\theta^{(2)}\|_1 + \mathbb{I}_{\mathcal{C}}(\theta) \tag{3}$$

8.2 ADMM Formulation

We reformulate this as a constrained optimization problem using the splitting:

$$\begin{aligned} \underset{\theta,z}{\text{minimize}} \quad & \frac{1}{2\gamma}\|\theta - w\|_2^2 + \sum_{k=1}^{2} \lambda_k\|z^{(k)}\|_1 && (4)\\ \text{subject to} \quad & D_{\text{block}}\theta = z && (5)\\ & \theta \in \mathcal{C} && (6) \end{aligned}$$

where $D_{\text{block}} = \text{diag}(D, D) \in \mathbb{R}^{2m\times 2n}$ is the block-diagonal difference matrix, and $z = (z^{(1)}, z^{(2)}) \in \mathbb{R}^{2m}$ are auxiliary variables representing the differences.

The augmented Lagrangian is:

$$\mathcal{L}_\rho(\theta, z, y) = \frac{1}{2\gamma}\|\theta - w\|_2^2 + \sum_{k=1}^{2} \lambda_k\|z^{(k)}\|_1 + y^T(D_{\text{block}}\theta - z) + \frac{\rho}{2}\|D_{\text{block}}\theta - z\|_2^2 \tag{7}$$

where $y \in \mathbb{R}^{2m}$ are dual variables and $\rho > 0$ is the penalty parameter.

8.3 ADMM Algorithm

The ADMM algorithm alternates between three steps:

Step 1: θ-update with constraint projection. Given current $(z^{(t)}, y^{(t)})$, solve:

$$\theta^{(t+1)} = \arg\min_{\theta\in\mathcal{C}} \left\{ \frac{1}{2\gamma}\|\theta - w\|_2^2 + \frac{\rho}{2}\|D_{\text{block}}\theta - z^{(t)} + \rho^{-1}y^{(t)}\|_2^2 \right\} \tag{8}$$

This is a constrained quadratic program. We solve it in two stages:

Stage 1a: Unconstrained solution. Ignoring the constraint $\theta \in \mathcal{C}$, the minimizer satisfies:

$$\left(\frac{1}{\gamma}I_{2n} + \rho D_{\text{block}}^T D_{\text{block}}\right)\tilde{\theta} = \frac{1}{\gamma}w + \rho D_{\text{block}}^T\left(z^{(t)} - \rho^{-1}y^{(t)}\right) \tag{9}$$

We solve this via sparse Cholesky decomposition:

$$A = \frac{1}{\gamma}I_{2n} + \rho D_{\text{block}}^T D_{\text{block}}, \quad A = LL^T, \quad \tilde{\theta} = A^{-1}b \tag{10}$$

Stage 1b: Constraint projection via PAVA. The unconstrained solution $\tilde{\theta} = (\tilde{\theta}^{(1)}, \tilde{\theta}^{(2)})$ may violate the ordering constraint. We project onto $\mathcal{C}$ pointwise using the Pool Adjacent Violators Algorithm (PAVA):

$$\theta_i^{(t+1)} = \Pi_{\mathcal{C}_i}(\tilde{\theta}_i^{(1)}, \tilde{\theta}_i^{(2)}), \quad i = 1, \ldots, n \tag{11}$$

where $\mathcal{C}_i = \{(a, b) : a \leq b\}$ and

$$\Pi_{\mathcal{C}_i}(a, b) = \begin{cases} (a, b) & \text{if } a \leq b \\ \left(\frac{a+b}{2}, \frac{a+b}{2}\right) & \text{if } a > b \end{cases} \tag{12}$$

This projection is optimal because the objective is separable in the components θ_i and the constraint is pointwise.

Step 2: z-update with quantile-specific soft-thresholding. Given $\theta^{(t+1)}$, solve:

$$z^{(t+1)} = \arg\min_z \left\{ \sum_{k=1}^{2} \lambda_k \|z^{(k)}\|_1 + \frac{\rho}{2} \|z - D_{\text{block}}\theta^{(t+1)} - \rho^{-1} y^{(t)}\|_2^2 \right\} \tag{13}$$

This separates by component and quantile. For each $j = 1, \ldots, m$ and $k \in \{1, 2\}$:

$$z_j^{(k,t+1)} = \mathcal{S}_{\lambda_k/\rho}\left([D\theta^{(k,t+1)}]_j + \rho^{-1} y_j^{(k,t)} \right) \tag{14}$$

where $\mathcal{S}_\kappa(x) = \text{sign}(x) \max(|x| - \kappa, 0)$ is the soft-thresholding operator.

Crucially, each quantile uses its own threshold λ_k/ρ, allowing different smoothness levels.

Step 3: Dual variable update.

$$y^{(t+1)} = y^{(t)} + \rho(D_{\text{block}}\theta^{(t+1)} - z^{(t+1)}) \tag{15}$$

8.4 Convergence Criteria

We monitor primal and dual residuals:

$$r_{\text{primal}} = \|D_{\text{block}}\theta^{(t+1)} - z^{(t+1)}\|_2 \tag{16}$$

$$r_{\text{dual}} = \rho \|D_{\text{block}}^T (z^{(t+1)} - z^{(t)})\|_2 \tag{17}$$

The algorithm terminates when both $r_{\text{primal}} < \epsilon$ and $r_{\text{dual}} < \epsilon$ for some tolerance $\epsilon > 0$.

8.5 Role in P-MALA

This ADMM solver computes the proximal operator $\text{prox}_{\gamma g}(w)$ required in the P-MALA proposal step. The key advantages are:

1. **Handles non-smooth ℓ_1 penalty** via soft-thresholding in the z-update
2. **Enforces hard constraints** via PAVA projection in the θ-update
3. **Quantile-specific smoothness** through separate λ_k parameters
4. **Computational efficiency** through sparse matrix operations

The ADMM solution need not converge to machine precision because the Metropolis-Hastings acceptance step in P-MALA corrects for inexact proximal operators, as established by Pereyra et al. (2016). In practice, we use tolerances $\epsilon \in [10^{-3}, 10^{-2}]$ and maximum iterations $T \in [100, 500]$.

The empirical failure and theoretical limitations led us to abandon the hierarchical approach in favor of the direct Gamma prior on λ presented in Section 1. While less theoretically elegant, this direct specification:

- Maintains conjugacy through $\lambda|\theta, y \sim \text{Gamma}(a, b + \|D\theta\|_1)$
- Avoids the pathological collapse of local variance parameters
- Provides computational efficiency through exact Gibbs sampling of λ
- Proves empirically robust across smooth, moderate, and high-frequency functions

9 ADMM alternative

This section describes a deterministic competitor to our proximal MCMC approach: an ADMM solver for the MAP estimator in smoothed quantile trend filtering. We first present ADMM in its *original (unscaled)* form and then derive the *scaled* form used in implementations.

9.1 Objective and splitting

Let $y \in \mathbb{R}^n$ be the observations and $\theta \in \mathbb{R}^n$ the latent signal (the τ-quantile curve). Let $D \in \mathbb{R}^{m \times n}$ denote the $(k+1)$st order discrete difference operator. We use the smooth approximation to the check loss

$$\tilde{\rho}_\tau(u) = \tau u + \varepsilon \log\bigl(1 + \exp(-u/\varepsilon)\bigr), \qquad \varepsilon > 0,$$

and define the smooth data-fit term

$$f(\theta) = \sum_{i=1}^{n} \tilde{\rho}_\tau(y_i - \theta_i).$$

The penalized estimator (MAP) solves

$$\min_{\theta \in \mathbb{R}^n} \; f(\theta) + \lambda \|D\theta\|_1, \qquad \lambda > 0.$$

To separate the nonsmooth ℓ_1 term, introduce an auxiliary variable

$$z = D\theta,$$

which yields the equivalent constrained problem

$$\min_{\theta \in \mathbb{R}^n,\; z \in \mathbb{R}^m} \; f(\theta) + \lambda \|z\|_1 \quad \text{s.t.} \quad D\theta - z = 0.$$

This splitting isolates the nonsmooth term in z while keeping $f(\theta)$ smooth.

9.2 Original (unscaled) ADMM

Introduce a Lagrange multiplier (dual variable) $y \in \mathbb{R}^m$ for the linear constraint $D\theta - z = 0$, and fix a penalty parameter $\rho > 0$. The *augmented Lagrangian* is

$$\mathcal{L}_\rho(\theta, z, y) = f(\theta) + \lambda \|z\|_1 + y^\top (D\theta - z) + \frac{\rho}{2} \|D\theta - z\|_2^2.$$

Compared to the ordinary Lagrangian, the additional quadratic term $\frac{\rho}{2}\|D\theta - z\|_2^2$ improves numerical stability and, for convex problems, yields strong convergence guarantees.

Given (θ^t, z^t, y^t), one *unscaled* ADMM iteration performs:

Step 1 (θ-update).

$$\theta^{t+1} = \arg\min_\theta \; f(\theta) + (y^t)^\top (D\theta - z^t) + \frac{\rho}{2} \|D\theta - z^t\|_2^2.$$

Step 2 (z-update).

$$z^{t+1} = \arg\min_z \; \lambda \|z\|_1 - (y^t)^\top z + \frac{\rho}{2} \|D\theta^{t+1} - z\|_2^2.$$

Step 3 (dual update).

$$y^{t+1} = y^t + \rho\,(D\theta^{t+1} - z^{t+1}).$$

The dual update increases (or decreases) the multiplier y in proportion to the constraint violation $D\theta - z$; at convergence we have $D\theta = z$.

9.3 Scaled ADMM and why it looks the way it does

In practice one typically works with the *scaled dual variable*

$$u^t = \frac{1}{\rho} y^t,$$

because it simplifies the algebra and the updates. Substituting $y = \rho u$ into the augmented Lagrangian gives

$$\mathcal{L}_\rho(\theta, z, \rho u) = f(\theta) + \lambda \|z\|_1 + \rho\, u^\top (D\theta - z) + \frac{\rho}{2} \|D\theta - z\|_2^2.$$

Now complete the square in the last two terms. Let $r = D\theta - z$. Then

$$\begin{aligned} \rho\, u^\top r + \frac{\rho}{2} \|r\|_2^2 &= \frac{\rho}{2} \Big(\|r\|_2^2 + 2u^\top r \Big) \\ &= \frac{\rho}{2} \Big(\|r + u\|_2^2 - \|u\|_2^2 \Big), \end{aligned}$$

since $\|r+u\|_2^2 = \|r\|_2^2 + 2u^\top r + \|u\|_2^2$. Therefore the *scaled augmented Lagrangian* can be written as

$$\mathcal{L}_\rho(\theta, z, u) = f(\theta) + \lambda \|z\|_1 + \frac{\rho}{2} \|D\theta - z + u\|_2^2 - \frac{\rho}{2} \|u\|_2^2.$$

The final term $-\frac{\rho}{2}\|u\|_2^2$ depends only on u (not on θ or z), so it does *not* change the θ- or z-minimizers. It appears solely because the squared form is algebraically identical to the original augmented Lagrangian after the change of variables $y = \rho u$.

With (θ^t, z^t, u^t), one *scaled* ADMM iteration becomes:

Step 1 (θ-update).

$$\theta^{t+1} = \arg\min_\theta \; f(\theta) + \frac{\rho}{2} \|D\theta - z^t + u^t\|_2^2.$$

Step 2 (z-update).

$$\begin{aligned} z^{t+1} &= \arg\min_z \; \lambda \|z\|_1 + \frac{\rho}{2} \|D\theta^{t+1} - z + u^t\|_2^2 \\ &= \mathsf{S}_{\lambda/\rho}\big(D\theta^{t+1} + u^t\big), \end{aligned}$$

where $\mathsf{S}_\kappa(\cdot)$ is componentwise soft-thresholding,

$$\big(\mathsf{S}_\kappa(x)\big)_j = \operatorname{sign}(x_j) \max\{|x_j| - \kappa, 0\}.$$

Step 3 (scaled dual update).

$$u^{t+1} = u^t + (D\theta^{t+1} - z^{t+1}).$$

This is the same dual ascent step as in the unscaled form, but expressed in $u = y/\rho$.

9.4 How the θ-update minimizes the augmented Lagrangian

Fix an iteration index t and treat (z^t, u^t) as constants. In the *scaled* formulation, the augmented Lagrangian is

$$\mathcal{L}_\rho(\theta, z, u) = f(\theta) + \lambda \|z\|_1 + \frac{\rho}{2} \|D\theta - z + u\|_2^2 - \frac{\rho}{2} \|u\|_2^2.$$

The ADMM θ-update is defined by

$$\theta^{t+1} = \arg\min_\theta \; \mathcal{L}_\rho(\theta, z^t, u^t).$$

To see why this reduces to the stated subproblem, observe that when $(z, u) = (z^t, u^t)$ is fixed, the terms

$$\lambda\|z^t\|_1 \quad \text{and} \quad -\frac{\rho}{2}\|u^t\|_2^2$$

do not depend on θ and hence do not affect the minimizer over θ. Therefore,

$$\theta^{t+1} = \arg\min_{\theta} \Big\{ f(\theta) + \frac{\rho}{2}\|D\theta - z^t + u^t\|_2^2 \Big\}.$$

First-order optimality condition. Define the θ-subproblem objective

$$\phi_t(\theta) = f(\theta) + \frac{\rho}{2}\|D\theta - z^t + u^t\|_2^2.$$

Since f is differentiable (smoothed check loss) and the quadratic term is differentiable, ϕ_t is differentiable. A necessary and sufficient condition for θ^{t+1} to minimize ϕ_t (convexity holds here) is

$$\nabla\phi_t(\theta^{t+1}) = 0.$$

Compute the gradient:

$$\begin{aligned}\nabla\phi_t(\theta) &= \nabla f(\theta) + \rho\nabla_\theta\Big(\frac{1}{2}\|D\theta - z^t + u^t\|_2^2\Big) \\ &= \nabla f(\theta) + \rho D^\top(D\theta - z^t + u^t).\end{aligned}$$

Thus θ^{t+1} satisfies

$$\nabla f(\theta^{t+1}) + \rho D^\top\big(D\theta^{t+1} - z^t + u^t\big) = 0.$$

Second-order structure and Newton's method. The Hessian of ϕ_t is

$$\nabla^2\phi_t(\theta) = \nabla^2 f(\theta) + \rho D^\top D,$$

which is positive definite under the present setup (the quadratic term contributes $\rho D^\top D$ and $\nabla^2 f(\theta) \succeq 0$), so ϕ_t is strongly convex and the minimizer is unique. Newton's method applies by solving at the current iterate θ:

$$\big(\nabla^2 f(\theta) + \rho D^\top D\big)\Delta = -\Big(\nabla f(\theta) + \rho D^\top(D\theta - z^t + u^t)\Big),$$

and updating $\theta \leftarrow \theta + \alpha\Delta$ (with line search), until convergence.

9.5 Stopping criteria

Define primal and dual residuals

$$\begin{aligned}r^{t+1} &= D\theta^{t+1} - z^{t+1}, \\ s^{t+1} &= \rho D^\top(z^{t+1} - z^t).\end{aligned}$$

We terminate when $\|r^{t+1}\|_2$ and $\|s^{t+1}\|_2$ fall below prescribed tolerances (absolute/relative), as in standard ADMM implementations.

10 Joint Estimation of Multiple Quantiles via Sequential Gibbs Sampling

10.1 Overview and Motivation

We extend the individual quantile estimation framework to estimate K quantiles $\tau_1 < \tau_2 < \cdots < \tau_K$ simultaneously. For concreteness, we focus on $K = 4$ quantiles, though the methodology extends naturally to arbitrary K.

The joint posterior distribution is:

$$\pi(\theta_1, \ldots, \theta_K, \lambda_1, \ldots, \lambda_K \mid y) \propto \prod_{i=1}^{K} [L(\theta_i \mid y, \tau_i) \cdot \pi(\theta_i \mid \lambda_i) \cdot \pi(\lambda_i)] \cdot \mathbb{I}[\theta_1 \leq \theta_2 \leq \cdots \leq \theta_K] \quad (18)$$

where $L(\theta_i \mid y, \tau_i) \propto \exp\left(-\sum_{j=1}^{n} \rho_{\tau_i}(y_j - \theta_{i,j})\right)$ is the asymmetric Laplace likelihood for quantile τ_i, and $\mathbb{I}[\cdot]$ is the indicator function enforcing the quantile ordering constraint.

Rather than updating all quantiles simultaneously in a high-dimensional block update (which would require solving a Kn-dimensional constrained optimization problem), we employ **sequential Gibbs sampling**, updating each quantile conditional on the others.

10.2 Sequential Gibbs Sampler Structure

The Gibbs sampler cycles through $K + 1$ update steps at each iteration t:

Algorithm 1 Sequential Gibbs Sampler for K Quantiles

Initialize: $\theta_i^{(0)}$, $\lambda_i^{(0)}$ for $i = 1, \ldots, K$
for $t = 1, 2, \ldots, T$ **do**
  for $i = 1, 2, \ldots, K$ **do**
    Sample $\theta_i^{(t)} \sim p(\theta_i \mid \theta_1^{(t)}, \ldots, \theta_{i-1}^{(t)}, \theta_{i+1}^{(t-1)}, \ldots, \theta_K^{(t-1)}, \lambda_i^{(t-1)}, y)$ via P-MALA
  end for
  for $i = 1, 2, \ldots, K$ **do**
    Sample $\lambda_i^{(t)} \sim \text{Gamma}(a, b + \|D\theta_i^{(t)}\|_1)$
  end for
end for

Note that within each outer iteration, quantile i is updated using the *newly updated* values $\theta_1^{(t)}, \ldots, \theta_{i-1}^{(t)}$ and the *previous* values $\theta_{i+1}^{(t-1)}, \ldots, \theta_K^{(t-1)}$. This creates an information cascade where updates propagate through the quantile hierarchy.

10.3 Full Conditional Distributions

10.3.1 First Quantile: $\theta_1 \mid \theta_2, \theta_3, \theta_4, \lambda_1, y$

The full conditional for θ_1 (the lowest quantile) is:

$$\pi(\theta_1 \mid \theta_2, \theta_3, \theta_4, \lambda_1, y) \propto \exp\left(-\sum_{j=1}^{n} \rho_{\tau_1}(y_j - \theta_{1,j}) - \lambda_1 \|D\theta_1\|_1\right) \cdot \mathbb{I}[\theta_1 \leq \theta_2] \quad (19)$$

Since $\theta_2, \theta_3, \theta_4$ are fixed during this update, the constraint $\theta_1 \leq \theta_2 \leq \theta_3 \leq \theta_4$ reduces to the simple box constraint $\theta_{1,j} \leq \theta_{2,j}$ for each $j = 1, \ldots, n$.

10.3.2 Middle Quantiles: $\theta_i \mid \theta_{-i}, \lambda_i, y$ for $i = 2, 3$

For intermediate quantiles, the full conditional is:

$$\pi(\theta_i \mid \theta_{-i}, \lambda_i, y) \propto \exp\left(-\sum_{j=1}^{n} \rho_{\tau_i}(y_j - \theta_{i,j}) - \lambda_i \|D\theta_i\|_1\right) \cdot \mathbb{I}[\theta_{i-1} \leq \theta_i \leq \theta_{i+1}] \quad (20)$$

The ordering constraint becomes a **two-sided box constraint**: $\theta_{i-1,j} \leq \theta_{i,j} \leq \theta_{i+1,j}$ for each location j.

10.3.3 Last Quantile: $\theta_K \mid \theta_1, \ldots, \theta_{K-1}, \lambda_K, y$

For the highest quantile:

$$\pi(\theta_K \mid \theta_1, \ldots, \theta_{K-1}, \lambda_K, y) \propto \exp\left(-\sum_{j=1}^{n} \rho_{\tau_K}(y_j - \theta_{K,j}) - \lambda_K \|D\theta_K\|_1\right) \cdot \mathbb{I}[\theta_{K-1} \leq \theta_K] \quad (21)$$

The constraint reduces to a one-sided box constraint (lower bound only): $\theta_{K-1,j} \leq \theta_{K,j}$.

10.4 P-MALA Update with Constrained Proximal Operators

Each full conditional update follows the standard P-MALA algorithm:

1. **Langevin diffusion step:** Generate proposal input

$$z_i = \theta_i^{\text{current}} - \delta \nabla f_i(\theta_i^{\text{current}}) + \sqrt{2\delta} \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I_n) \quad (22)$$

   where $f_i(\theta) = \sum_{j=1}^{n} \tilde{\rho}_{\tau_i}(y_j - \theta_j; \varepsilon)$ is the smoothed check loss.

2. **Proximal operator step:** Compute

$$\theta_i^{\text{proposal}} = \text{prox}_{\gamma g_i}(z_i) \quad (23)$$

   where $g_i(\theta) = \lambda_i \|D\theta\|_1 + I_{\mathcal{C}_i}(\theta)$ with $\mathcal{C}_i$ being the appropriate box constraint set.

3. **Metropolis-Hastings step:** Accept or reject via standard P-MALA acceptance ratio.

The key computational challenge is solving the constrained proximal operator:

$$\text{prox}_{\gamma g_i}(z_i) = \arg\min_{\theta_i} \frac{1}{2\gamma} \|\theta_i - z_i\|_2^2 + \lambda_i \|D\theta_i\|_1 + I_{\mathcal{C}_i}(\theta_i) \quad (24)$$

We solve this using the Alternating Direction Method of Multipliers (ADMM).

10.5 ADMM Formulation for Constrained Proximal Operators

We present the ADMM formulation for quantile i with two-sided box constraints $\ell \leq \theta_i \leq u$ (the one-sided cases are special cases with $\ell = -\infty$ or $u = +\infty$).

10.5.1 Consensus Formulation

Introduce auxiliary variables:

- $v_i \in \mathbb{R}^m$ for the trend filtering penalty (where $m = n - k - 1$)
- $w_i \in \mathbb{R}^n$ for the box constraint

Reformulate problem (24) as:

$$\begin{aligned} \min_{\theta_i, v_i, w_i} \quad & \frac{1}{2\gamma} \|w_i - z_i\|_2^2 + \lambda_i \|v_i\|_1 \\ \text{subject to} \quad & D\theta_i = v_i \\ & \theta_i = w_i \\ & \ell \leq w_i \leq u \end{aligned} \quad (25)$$

10.5.2 Augmented Lagrangian

The augmented Lagrangian for (25) is:

$$\begin{aligned}\mathcal{L}_\rho(\theta_i, v_i, w_i, u_i, y_i) = \frac{1}{2\gamma}\|w_i - z_i\|_2^2 + \lambda_i\|v_i\|_1 \\ + \frac{\rho_1}{2}\|D\theta_i - v_i + u_i\|_2^2 \\ + \frac{\rho_2}{2}\|\theta_i - w_i + y_i\|_2^2\end{aligned} \tag{26}$$

where $u_i \in \mathbb{R}^m$ and $y_i \in \mathbb{R}^n$ are scaled dual variables (Lagrange multipliers divided by penalty parameters), and $\rho_1, \rho_2 > 0$ are penalty parameters.

10.5.3 ADMM Iterations

The ADMM algorithm alternates between minimizing $\mathcal{L}_\rho$ over primal variables and updating dual variables:

θ_i-update (Unconstrained Quadratic Program):

$$\theta_i^{(k+1)} = \arg\min_{\theta_i} \frac{\rho_1}{2}\|D\theta_i - v_i^{(k)} + u_i^{(k)}\|_2^2 + \frac{\rho_2}{2}\|\theta_i - w_i^{(k)} + y_i^{(k)}\|_2^2 \tag{27}$$

This is an unconstrained quadratic program. Taking the gradient and setting to zero:

$$\rho_1 D^\top(D\theta_i - v_i^{(k)} + u_i^{(k)}) + \rho_2(\theta_i - w_i^{(k)} + y_i^{(k)}) = 0 \tag{28}$$

Rearranging:

$$\underbrace{(\rho_1 D^\top D + \rho_2 I)}_{A_i}\theta_i = \rho_1 D^\top(v_i^{(k)} - u_i^{(k)}) + \rho_2(w_i^{(k)} - y_i^{(k)}) \tag{29}$$

The system matrix $A_i = \rho_1 D^\top D + \rho_2 I$ is:

- Symmetric positive definite (since $\rho_2 > 0$)
- Sparse and banded (inheriting structure from $D^\top D$)
- Fixed across all ADMM iterations

Computational Strategy: Precompute the sparse Cholesky factorization $A_i = L_i L_i^\top$ once before the ADMM loop. Then solve (29) via forward and backward substitution at $O(n)$ cost per iteration.

v_i-update (Soft-Thresholding):

$$v_i^{(k+1)} = \arg\min_{v_i} \lambda_i\|v_i\|_1 + \frac{\rho_1}{2}\|D\theta_i^{(k+1)} - v_i + u_i^{(k)}\|_2^2 \tag{30}$$

This is separable across components of v_i, with closed-form solution given by the soft-thresholding operator:

$$v_{i,j}^{(k+1)} = \mathcal{S}_{\lambda_i/\rho_1}((D\theta_i^{(k+1)})_j + u_{i,j}^{(k)}), \quad j = 1, \ldots, m \tag{31}$$

where $\mathcal{S}_\alpha(x) = \text{sign}(x) \cdot \max(|x| - \alpha, 0)$ is the soft-thresholding operator.

w_i-update (Box-Constrained Quadratic Program):

$$w_i^{(k+1)} = \arg\min_{w_i} \frac{1}{2\gamma}\|w_i - z_i\|_2^2 + \frac{\rho_2}{2}\|\theta_i^{(k+1)} - w_i + y_i^{(k)}\|_2^2 \quad \text{subject to } \ell \leq w_i \leq u \tag{32}$$

Taking the gradient of the unconstrained objective:

$$\frac{1}{\gamma}(w_i - z_i) + \rho_2(w_i - \theta_i^{(k+1)} - y_i^{(k)}) = 0 \tag{33}$$

Solving for w_i:

$$\tilde{w}_i = \frac{\frac{1}{\gamma}z_i + \rho_2(\theta_i^{(k+1)} + y_i^{(k)})}{\frac{1}{\gamma} + \rho_2} \tag{34}$$

The constrained solution is the Euclidean projection of $\tilde{w}_i$ onto the box $[\ell, u]$:

$$w_i^{(k+1)} = \Pi_{[\ell,u]}(\tilde{w}_i) = \max(\ell, \min(\tilde{w}_i, u)) \tag{35}$$

where the max and min operations are applied elementwise.

This projection is the **proximal operator** of the indicator function of the box constraint set:

$$\Pi_{[\ell,u]}(x) = \text{prox}_{I_{[\ell,u]}}(x) = \arg\min_{w \in [\ell,u]} \frac{1}{2}\|w - x\|_2^2 \tag{36}$$

Dual Variable Updates: Update the scaled dual variables via gradient ascent on the dual function:

$$u_i^{(k+1)} = u_i^{(k)} + D\theta_i^{(k+1)} - v_i^{(k+1)} \tag{37}$$

$$y_i^{(k+1)} = y_i^{(k)} + \theta_i^{(k+1)} - w_i^{(k+1)} \tag{38}$$

10.5.4 Convergence Criteria

Monitor primal and dual residuals:

$$r_{\text{primal}}^{(1)} = \|D\theta_i^{(k)} - v_i^{(k)}\|_2 \quad \text{(consensus on } D\theta_i = v_i) \tag{39}$$

$$r_{\text{primal}}^{(2)} = \|\theta_i^{(k)} - w_i^{(k)}\|_2 \quad \text{(consensus on } \theta_i = w_i) \tag{40}$$

$$r_{\text{dual}}^{(1)} = \|\rho_1 D^\top (v_i^{(k)} - v_i^{(k-1)})\|_2 \tag{41}$$

$$r_{\text{dual}}^{(2)} = \|\rho_2 (w_i^{(k)} - w_i^{(k-1)})\|_2 \tag{42}$$

Terminate when:

$$\max\{r_{\text{primal}}^{(1)}, r_{\text{primal}}^{(2)}, r_{\text{dual}}^{(1)}, r_{\text{dual}}^{(2)}\} < \epsilon_{\text{tol}} \tag{43}$$

where typical tolerance is $\epsilon_{\text{tol}} = 10^{-4}$.

10.6 Specific Constraint Sets for Four Quantiles

For $K = 4$ quantiles $\tau_1 < \tau_2 < \tau_3 < \tau_4$, the box constraints for each update are:

Quantile	Constraint Set $\mathcal{C}_i$	Box Bounds $[\ell, u]$
θ_1	$\{\theta : \theta \leq \theta_2\}$	$[-\infty, \theta_2^{\text{current}}]$
θ_2	$\{\theta : \theta_1 \leq \theta \leq \theta_3\}$	$[\theta_1^{\text{new}}, \theta_3^{\text{current}}]$
θ_3	$\{\theta : \theta_2 \leq \theta \leq \theta_4\}$	$[\theta_2^{\text{new}}, \theta_4^{\text{current}}]$
θ_4	$\{\theta : \theta_3 \leq \theta\}$	$[\theta_3^{\text{new}}, +\infty]$

Table 2: Box constraint sets for each quantile in the sequential Gibbs sampler.

Note that quantiles 2 and 3 use *newly updated* values for their lower bounds (creating the information cascade) and *current* values for upper bounds.

10.7 The Role of the Auxiliary Variable w : A Complete Explanation

The introduction of the auxiliary variable w in our ADMM formulation is not merely a technical convenienceâit is essential for computational tractability. This subsection provides a complete explanation of why w is needed, how it transforms the problem, and why it enables efficient computation.

10.7.1 The Fundamental Challenge

Recall that each quantile update in the sequential Gibbs sampler requires computing a constrained proximal operator:

$$\text{prox}_{\gamma g_i}(z_i) = \arg\min_{\theta_i} \frac{1}{2\gamma}\|\theta_i - z_i\|_2^2 + \lambda_i \|D\theta_i\|_1 + I_{\mathcal{C}_i}(\theta_i) \tag{44}$$

where z_i is the input from the Langevin diffusion step, $\mathcal{C}_i$ is the box constraint set (e.g., $\{\theta : \ell \leq \theta \leq u\}$), and $I_{\mathcal{C}_i}$ is the indicator function:

$$I_{\mathcal{C}_i}(\theta) = \begin{cases} 0 & \text{if } \theta \in \mathcal{C}_i \\ +\infty & \text{if } \theta \notin \mathcal{C}_i \end{cases} \tag{45}$$

This problem presents two computational challenges:

1. The ℓ_1 penalty $\|D\theta_i\|_1$ is non-smooth, requiring iterative optimization (ADMM)
2. The box constraints $\theta \in \mathcal{C}_i$ must be enforced during optimization

The critical question is: *How do we handle both challenges simultaneously in a computationally efficient manner?*

10.7.2 Naive Approach: Direct ADMM Without w

A natural first attempt is to formulate ADMM directly on problem (44).

ADMM Formulation. Introduce auxiliary variable $v_i = D\theta_i$ to handle the ℓ_1 penalty:

$$\begin{aligned} \underset{\theta_i, v_i}{\text{minimize}} \quad & \frac{1}{2\gamma}\|\theta_i - z_i\|_2^2 + \lambda_i \|v_i\|_1 \\ \text{subject to} \quad & D\theta_i = v_i \\ & \ell \leq \theta_i \leq u \end{aligned} \tag{46}$$

The augmented Lagrangian is:

$$\mathcal{L}_\rho(\theta_i, v_i, u_i) = \frac{1}{2\gamma}\|\theta_i - z_i\|_2^2 + \lambda_i \|v_i\|_1 + \frac{\rho}{2}\|D\theta_i - v_i + u_i\|_2^2 \tag{47}$$

where $u_i \in \mathbb{R}^m$ is the scaled dual variable and $\rho > 0$ is a penalty parameter.

The v_i-update This update is:

$$v_i^{(k+1)} = \arg\min_{v_i} \lambda_i \|v_i\|_1 + \frac{\rho}{2}\|D\theta_i^{(k)} - v_i + u_i^{(k)}\|_2^2 \tag{48}$$

The objective is separable across components, with closed-form solution via soft-thresholding:

$$[v_i^{(k+1)}]_j = \mathcal{S}_{\lambda_i/\rho}([D\theta_i^{(k)} + u_i^{(k)}]_j), \quad j = 1, \ldots, m \tag{49}$$

where $\mathcal{S}_\alpha(x) = \text{sign}(x) \cdot \max(|x| - \alpha, 0)$ is the soft-thresholding operator.
Computational cost: $O(m)$ where $m = n - k - 1$.

The θ_i-update This is where the problem arises:

$$\theta_i^{(k+1)} = \arg\min_{\theta_i} \frac{1}{2\gamma}\|\theta_i - z_i\|_2^2 + \frac{\rho}{2}\|D\theta_i - v_i^{(k)} + u_i^{(k)}\|_2^2 \quad \text{subject to } \ell \le \theta_i \le u \tag{50}$$

Expanding the objective. Let us expand both terms carefully.

First term:

$$\begin{aligned}\frac{1}{2\gamma}\|\theta_i - z_i\|_2^2 &= \frac{1}{2\gamma}(\theta_i - z_i)^T(\theta_i - z_i) \\ &= \frac{1}{2\gamma}\theta_i^T\theta_i - \frac{1}{\gamma}\theta_i^T z_i + \frac{1}{2\gamma}z_i^T z_i\end{aligned} \tag{51}$$

Second term:

$$\begin{aligned}\frac{\rho}{2}\|D\theta_i - v_i^{(k)} + u_i^{(k)}\|_2^2 &= \frac{\rho}{2}(D\theta_i - v_i^{(k)} + u_i^{(k)})^T(D\theta_i - v_i^{(k)} + u_i^{(k)}) \\ &= \frac{\rho}{2}\theta_i^T D^T D\theta_i - \rho\theta_i^T D^T(v_i^{(k)} - u_i^{(k)}) + \text{const}\end{aligned} \tag{52}$$

Combined objective (dropping constants):

$$\frac{1}{2}\theta_i^T \underbrace{\left(\frac{1}{\gamma}I + \rho D^T D\right)}_{A_i}\theta_i - \theta_i^T \underbrace{\left(\frac{1}{\gamma}z_i + \rho D^T(v_i^{(k)} - u_i^{(k)})\right)}_{b_i} \tag{53}$$

Define:

$$\boxed{A_i = \frac{1}{\gamma}I + \rho D^T D} \quad \text{and} \quad \boxed{b_i = \frac{1}{\gamma}z_i + \rho D^T(v_i^{(k)} - u_i^{(k)})} \tag{54}$$

So problem (50) becomes:

$$\theta_i^{(k+1)} = \arg\min_{\theta_i \in [\ell, u]} \frac{1}{2}\theta_i^T A_i\theta_i - \theta_i^T b_i \tag{55}$$

Finding the unconstrained minimizer. Ignoring constraints temporarily, take the gradient:

$$\nabla_{\theta_i}\left[\frac{1}{2}\theta_i^T A_i\theta_i - \theta_i^T b_i\right] = A_i\theta_i - b_i = 0 \tag{56}$$

Therefore:

$$\tilde{\theta}_i = A_i^{-1}b_i \tag{57}$$

This is straightforward to compute because $A_i = \frac{1}{\gamma}I + \rho D^T D$ is:

- Symmetric positive definite
- Sparse and banded (inheriting structure from $D^T D$)
- Can be factored once: $A_i = L_i L_i^T$ (sparse Cholesky)
- Each solve costs $O(n)$ using the factorization

So finding $\tilde{\theta}_i$ is computationally easy.

Why we cannot simply project. Having computed the unconstrained minimizer $\tilde{\theta}_i$, one might hope to enforce constraints by simple clipping:

$$\theta_i^{(k+1)} \stackrel{?}{=} \max(\ell, \min(\tilde{\theta}_i, u)) \quad \text{[elementwise]} \tag{58}$$

This is incorrect.

The fundamental issue. Clipping computes:

$$\arg\min_{\theta_i \in [\ell,u]} \|\theta_i - \tilde{\theta}_i\|_2^2 = \arg\min_{\theta_i \in [\ell,u]} \theta_i^T \mathbf{I} \theta_i - 2\theta_i^T \tilde{\theta}_i \tag{59}$$

This minimizes Euclidean distance to $\tilde{\theta}_i$, which corresponds to a quadratic form with matrix I (the identity).

However, what we need is:

$$\arg\min_{\theta_i \in [\ell,u]} \theta_i^T \mathbf{A_i} \theta_i - 2\theta_i^T b_i \tag{60}$$

where $A_i = \frac{1}{\gamma} I + \rho D^T D \neq \alpha I$ for any scalar α (unless $D^T D = 0$, which is not the case).
These are fundamentally different optimization problems.

What is actually required. To solve (55) correctly, one needs:

1. **Projected gradient descent:** Alternate gradient steps with projection

   repeat
   $\theta_i \leftarrow \theta_i - \alpha(A_i \theta_i - b_i)$ ▷ Gradient step
   $\theta_i \leftarrow \max(\ell, \min(\theta_i, u))$ ▷ Project onto box
   until convergence

   Requires many iterations (typically 50-200) to converge.

2. **Active set methods:** Identify which constraints are active (binding) at the solution, solve the corresponding KKT system. Complex algorithm, difficult to implement efficiently.

3. **Interior point methods:** Add barrier functions for constraints, solve a sequence of unconstrained problems. Even more complex.

10.7.3 The Solution: Introducing Auxiliary Variable w

The key insight is to **decouple** the two difficult aspects of the problem:

1. The trend filtering structure (encoded in $D^T D$)
2. The box constraints

We achieve this by introducing a second auxiliary variable w with the constraint $w = \theta$.

Modified ADMM formulation. Introduce **two** auxiliary variables:

- $v_i = D\theta_i$ (for the ℓ_1 penalty, as before)
- $w_i = \theta_i$ (new variable for the box constraint)

Crucially, move the data fidelity term from θ_i to w_i:

$$\begin{aligned} \underset{\theta_i, v_i, w_i}{\text{minimize}} \quad & \frac{1}{2\gamma} \|w_i - z_i\|_2^2 + \lambda_i \|v_i\|_1 \\ \text{subject to} \quad & D\theta_i = v_i \\ & \theta_i = w_i \\ & \ell \leq w_i \leq u \end{aligned} \tag{61}$$

Notice:

- The data term $\frac{1}{2\gamma} \| \cdot - z_i\|^2$ depends on w_i, not θ_i
- The box constraints apply to w_i, not θ_i
- The variable θ_i appears only in the equality constraints

The augmented Lagrangian is:

$$\begin{aligned}\mathcal{L}_\rho(\theta_i, v_i, w_i, u_{1,i}, u_{2,i}) = &\frac{1}{2\gamma}\|w_i - z_i\|_2^2 + \lambda_i\|v_i\|_1 \\ &+ \frac{\rho_1}{2}\|D\theta_i - v_i + u_{1,i}\|_2^2 \\ &+ \frac{\rho_2}{2}\|\theta_i - w_i + u_{2,i}\|_2^2\end{aligned} \tag{62}$$

where $u_{1,i} \in \mathbb{R}^m$ and $u_{2,i} \in \mathbb{R}^n$ are scaled dual variables, and $\rho_1, \rho_2 > 0$ are penalty parameters.

The three ADMM updates. The ADMM algorithm alternates:

1. θ_i-update (now unconstrained):

$$\theta_i^{(k+1)} = \arg\min_{\theta_i} \frac{\rho_1}{2}\|D\theta_i - v_i^{(k)} + u_{1,i}^{(k)}\|_2^2 + \frac{\rho_2}{2}\|\theta_i - w_i^{(k)} + u_{2,i}^{(k)}\|_2^2 \tag{63}$$

No box constraints appear! This is an unconstrained quadratic program. Taking the gradient and setting to zero:

$$\rho_1 D^T(D\theta_i - v_i^{(k)} + u_{1,i}^{(k)}) + \rho_2(\theta_i - w_i^{(k)} + u_{2,i}^{(k)}) = 0 \tag{64}$$

Rearranging:

$$\underbrace{(\rho_1 D^T D + \rho_2 I)}_{A_i^{\text{new}}}\theta_i = \rho_1 D^T(v_i^{(k)} - u_{1,i}^{(k)}) + \rho_2(w_i^{(k)} - u_{2,i}^{(k)}) \tag{65}$$

Solve this sparse banded system via precomputed Cholesky factorization. Cost: $O(n)$.

2. v_i-update (soft-thresholding, unchanged):

$$v_i^{(k+1)} = \arg\min_{v_i} \lambda_i\|v_i\|_1 + \frac{\rho_1}{2}\|D\theta_i^{(k+1)} - v_i + u_{1,i}^{(k)}\|_2^2 \tag{66}$$

Solution:

$$[v_i^{(k+1)}]_j = \mathcal{S}_{\lambda_i/\rho_1}([D\theta_i^{(k+1)} + u_{1,i}^{(k)}]_j), \quad j = 1, \ldots, m \tag{67}$$

Cost: $O(m) \approx O(n)$.

3. w_i-update (the crucial step):

$$w_i^{(k+1)} = \arg\min_{w_i} \frac{1}{2\gamma}\|w_i - z_i\|_2^2 + \frac{\rho_2}{2}\|\theta_i^{(k+1)} - w_i + u_{2,i}^{(k)}\|_2^2 \quad \text{subject to } \ell \le w_i \le u \tag{68}$$

This is where the magic happens.

10.7.4 Why the w_i-Update is Computationally Trivial

Let us work through the w_i-update in complete detail, showing why it reduces to simple elementwise clipping.

Step 1: Expand the objective function. Starting from (68), drop iteration superscripts for clarity:

$$\min_{w_i \in [\ell, u]} \frac{1}{2\gamma}\|w_i - z_i\|_2^2 + \frac{\rho_2}{2}\|\theta_i - w_i + u_{2,i}\|_2^2 \tag{69}$$

First term:

$$\begin{aligned}\frac{1}{2\gamma}\|w_i - z_i\|_2^2 &= \frac{1}{2\gamma}(w_i - z_i)^T(w_i - z_i) \\ &= \frac{1}{2\gamma}w_i^T w_i - \frac{1}{\gamma}w_i^T z_i + \frac{1}{2\gamma}z_i^T z_i\end{aligned} \tag{70}$$

Second term:

$$\frac{\rho_2}{2}\|\theta_i - w_i + u_{2,i}\|_2^2 = \frac{\rho_2}{2}(\theta_i + u_{2,i} - w_i)^T(\theta_i + u_{2,i} - w_i)$$
$$= \frac{\rho_2}{2}w_i^T w_i - \rho_2 w_i^T(\theta_i + u_{2,i}) + \frac{\rho_2}{2}(\theta_i + u_{2,i})^T(\theta_i + u_{2,i}) \tag{71}$$

Combine both terms (drop constants independent of w_i):

$$\left(\frac{1}{2\gamma} + \frac{\rho_2}{2}\right) w_i^T w_i - w_i^T\left[\frac{1}{\gamma}z_i + \rho_2(\theta_i + u_{2,i})\right] \tag{72}$$

Step 2: Define α and c. Let:

$$\boxed{\alpha = \frac{1}{\gamma} + \rho_2} \quad \text{and} \quad \boxed{c = \frac{1}{\gamma}z_i + \rho_2(\theta_i + u_{2,i})} \tag{73}$$

Then the objective becomes:

$$\frac{1}{2}w_i^T(\alpha I)w_i - w_i^T c \tag{74}$$

Critical observation: The matrix is αI where α is a positive scalar. There is **no** $D^T D$ **term**âthe matrix is proportional to the identity.

Step 3: Find the unconstrained minimizer. Ignoring constraints, take the gradient of (74):

$$\nabla_{w_i}\left[\frac{1}{2}w_i^T(\alpha I)w_i - w_i^T c\right] = \alpha I w_i - c = \alpha w_i - c = 0 \tag{75}$$

Therefore:

$$\tilde{w}_i = \frac{c}{\alpha} = \frac{\frac{1}{\gamma}z_i + \rho_2(\theta_i + u_{2,i})}{\frac{1}{\gamma} + \rho_2} \tag{76}$$

This is a **weighted average** of z_i and $\theta_i + u_{2,i}$. It is trivial to computeâjust elementwise arithmetic in $O(n)$ time.

Step 4: Solve the constrained problem via projection. The constrained optimization problem is:

$$\arg\min_{w_i \in [\ell,u]} \frac{1}{2}w_i^T(\alpha I)w_i - w_i^T c \tag{77}$$

Since $\alpha > 0$, we can divide the entire objective by α without changing the minimizer:

$$\arg\min_{w_i \in [\ell,u]} \frac{1}{2}w_i^T w_i - w_i^T \frac{c}{\alpha} \tag{78}$$

Substituting $\tilde{w}_i = c/\alpha$:

$$= \arg\min_{w_i \in [\ell,u]} \frac{1}{2}w_i^T w_i - w_i^T \tilde{w}_i \tag{79}$$

Step 5: Complete the square. Recall the identity:

$$\|w_i - \tilde{w}_i\|^2 = w_i^T w_i - 2w_i^T\tilde{w}_i + \tilde{w}_i^T\tilde{w}_i \tag{80}$$

Therefore:

$$\frac{1}{2}w_i^T w_i - w_i^T\tilde{w}_i = \frac{1}{2}\|w_i - \tilde{w}_i\|^2 - \frac{1}{2}\|\tilde{w}_i\|^2 \tag{81}$$

The term $-\frac{1}{2}\|\tilde{w}_i\|^2$ is constant (independent of w_i), so:

$$\arg\min_{w_i \in [\ell,u]} \frac{1}{2}w_i^T w_i - w_i^T\tilde{w}_i = \arg\min_{w_i \in [\ell,u]} \frac{1}{2}\|w_i - \tilde{w}_i\|^2 \tag{82}$$

Step 6: Apply the closed-form projection formula. The Euclidean projection onto a box has a simple closed form:

$$\Pi_{[\ell,u]}(\tilde{w}_i) = \max(\ell, \min(\tilde{w}_i, u)) \quad \text{[elementwise]} \tag{83}$$

In components:

$$[\Pi_{[\ell,u]}(\tilde{w}_i)]_j = \begin{cases} \ell_j & \text{if } \tilde{w}_{i,j} < \ell_j \\ \tilde{w}_{i,j} & \text{if } \ell_j \leq \tilde{w}_{i,j} \leq u_j \\ u_j & \text{if } \tilde{w}_{i,j} > u_j \end{cases} \tag{84}$$

Computational cost: $O(n)$ elementwise operations.
Therefore, the solution to the w_i-update is:

$$\boxed{w_i^{(k+1)} = \max\left(\ell, \min\left(\tilde{w}_i, u\right)\right)} \tag{85}$$

where $\tilde{w}_i$ is given by (76).

10.7.5 The Key Mathematical Principle

The reason projection works for the w_i-update but not for the θ_i-update (without w) is a fundamental property of quadratic forms.

Theorem 1 (Projection Solves Constrained Minimization with Identity Matrix)**.** *Let C be a convex set, $\tilde{x} \in \mathbb{R}^n$, and $\alpha > 0$ a positive scalar. Then:*

$$\arg\min_{x \in C} \frac{1}{2} x^T (\alpha I) x - x^T c = \Pi_C(c/\alpha) \tag{86}$$

where $\Pi_C(\cdot)$ denotes Euclidean projection onto C.

Proof. Dividing the objective by $\alpha > 0$ (which doesn't change the minimizer):

$$\arg\min_{x \in C} \frac{1}{2} x^T x - x^T (c/\alpha) = \arg\min_{x \in C} \frac{1}{2} \|x - c/\alpha\|^2 = \Pi_C(c/\alpha) \tag{87}$$

The last equality is the definition of Euclidean projection. □

Crucial point: This theorem requires the matrix to be αI (a scalar times the identity). It does **not** hold for general positive definite matrices A.
In our problem:

- w_i-update: matrix is $\alpha I = (\frac{1}{\gamma} + \rho_2) I \Rightarrow$ projection works
- θ_i-update (without w): matrix is $A = \frac{1}{\gamma} I + \rho D^T D \neq \alpha I \Rightarrow$ projection fails

10.7.6 Connection to Proximal Operators

The equivalence between projection and the w_i-update can also be understood through the language of proximal operators.

Indicator functions. The indicator function of a set C is:

$$I_C(x) = \begin{cases} 0 & \text{if } x \in C \\ +\infty & \text{if } x \notin C \end{cases} \tag{88}$$

Adding $I_C(x)$ to an objective function enforces the constraint $x \in C$ (since the objective becomes infinite outside C).

Proximal operator definition. The proximal operator of a function h is:

$$\text{prox}_h(x) = \arg\min_z \left[\frac{1}{2}\|z - x\|^2 + h(z)\right] \tag{89}$$

Theorem 2 (Proximal Operator of Indicator Function)**.** *For any closed convex set C:*

$$prox_{I_C}(x) = \Pi_C(x) \tag{90}$$

That is, the proximal operator of the indicator function equals Euclidean projection.

Proof. By definition:

$$\text{prox}_{I_C}(x) = \arg\min_z \left[\frac{1}{2}\|z - x\|^2 + I_C(z)\right] \tag{91}$$

If $z \notin C$, then $I_C(z) = +\infty$ and z cannot be optimal. If $z \in C$, then $I_C(z) = 0$, so:

$$\text{prox}_{I_C}(x) = \arg\min_{z \in C} \frac{1}{2}\|z - x\|^2 = \Pi_C(x) \tag{92}$$

□

Application to our problem. The w_i-update can be written as:

$$\arg\min_{w_i} \left[\frac{1}{2}\|w_i - \tilde{w}_i\|^2 + I_{[\ell,u]}(w_i)\right] = \text{prox}_{I_{[\ell,u]}}(\tilde{w}_i) = \Pi_{[\ell,u]}(\tilde{w}_i) \tag{93}$$

This is why equation (36) in the text states:

$$\Pi_{[\ell,u]}(x) = \text{prox}_{I_{[\ell,u]}}(x) = \arg\min_{w \in [\ell,u]} \frac{1}{2}\|w - x\|_2^2 \tag{94}$$

These three expressions are equivalent ways of writing the same operation.

11 Theory to prove

We consider observations

$$Y_i = f_0(x_i) + \varepsilon_i, \qquad \varepsilon_i \sim N(0, \sigma^2), \quad i = 1, \ldots, n,$$

and define the shifted regression effects

$$\theta_{0,i} = f_0(x_i) + \Phi^{-1}(\tau)\sigma,$$
$$\theta_i = f(x_i) + \Phi^{-1}(\tau)\sigma,$$

corresponding respectively to the true function f_0 and an arbitrary candidate f. The goal is to establish posterior contraction at rate ε_n under the empirical L_2 norm

$$d_n(\theta, \theta_0) = \left(\frac{1}{n}\sum_{i=1}^{n}(\theta_i - \theta_{0,i})^2\right)^{1/2}.$$

Our target statement is

$$\Pi(d_n(\theta, \theta_0) \geq M_n\varepsilon_n \,|\, X_n) \longrightarrow 0 \qquad \text{in } P_0^n\text{-probability},$$

for any sequence $M_n \to \infty$. To apply a standard posterior contraction theorem, we construct an appropriate sieve Θ_n and verify the three canonical conditions: an entropy bound, a prior mass bound, and a prior concentration bound around θ_0.

Step 1. Construction of the sieve. We introduce a restriction set

$$\Theta_n = \big\{\theta : \text{regularity, variation, and magnitude constraints depending on } \varepsilon_n\big\},$$

to be chosen so that (i) it contains a sufficiently large neighborhood of θ_0; (ii) its complement receives exponentially small prior mass; and (iii) the set

$$\{\theta \in \Theta_n : d_n(\theta, \theta_0) < \varepsilon\}$$

has manageable complexity for all $\varepsilon > \varepsilon_n$.

The entropy requirement is

$$\sup_{\varepsilon > \varepsilon_n} \log N\Big(\frac{\varepsilon}{36},\ \{\theta \in \Theta_n : d_n(\theta, \theta_0) < \varepsilon\},\ d_n\Big)\ \le\ n\varepsilon_n^2.$$

Because the set $\{\theta \in \Theta_n : d_n(\theta, \theta_0) < \varepsilon\}$ is bounded, this condition is achievable once Θ_n controls the total variation and/or magnitude of θ.

Step 2. Prior mass of the sieve complement. A necessary condition is

$$\frac{\Pi(\Theta_n^c)}{\Pi(B_n(\theta_0, \varepsilon_n))} = o\Big(e^{-n\varepsilon_n^2}\Big),$$

where $B_n(\theta_0, \varepsilon_n)$ denotes the d_n ball around θ_0 of radius ε_n. It suffices to verify

1) $\Pi(\Theta_n^c) \le \exp(-n\varepsilon_n^2)$, (negligible mass outside the sieve,

2) $\Pi\big(B_n(\theta_0, \varepsilon_n)\big) \ge \exp(-n\varepsilon_n^2)$, (adequate mass near θ_0).

For 1) we must determine precise properties of both the sieve Θ_n and the prior. 2) requires precise control of the TV prior around θ_0, and specifically lower bounds on the prior density along directions close to θ_0 in $L_{2,n}$.

Step 3. Key properties of the TV prior. For the Laplace prior on discrete differences,

$$\Pi(\theta) \propto \exp(-\lambda \frac{n^k}{k!} \|D\theta\|_1),\ k = 3$$

the following ingredients are essential:

1) Structural properties: log–concavity, subgradient form, and the piecewise polynomial behavior;

2) Bounds on KL and variance terms V_k in terms of $d_n(\theta, \theta_0)$;

3) Local prior thickness: $\Pi(\|\theta - \theta_0\| \le \varepsilon_n) \gtrsim \exp(-n\varepsilon_n^2)$.

The last condition ensures that the prior allocates sufficient mass to neighborhoods of θ_0 compatible with the empirical L_2 ball, and typically requires controlling $\|D\theta - D\theta_0\|_1$ on sets where $d_n(\theta, \theta_0)$ is small.

Step 4. Additional annulus bound. To complete the testing argument, we also need for every integer $j \ge 1$,

$$\Pi\left(\theta \in \Theta_n : j\varepsilon_n < d_n(\theta, \theta_0) \le 2j\varepsilon_n\right)\ \le\ \exp\left(n\varepsilon_n^2 j^2\right),$$

which controls prior mass on annuli and is standard in the Ghosal–Ghosh–van der Vaart approach. This is typically handled after establishing the main two prior bounds.

Relationship with function space distance. Since

$$\begin{aligned} d_n(\theta, \theta_0) = \ell_{2,n}(\theta, \theta_0) &= \left(\frac{1}{n}\sum_{i=1}^n (\theta_i - \theta_{0,i})^2\right)^{1/2} \\ &= \left(\frac{1}{n}\sum_{i=1}^n (f(x_i) - f_0(x_i))^2\right)^{1/2} = L_{2,n}(f, f_0), \end{aligned}$$

the posterior contraction of θ is equivalent to contraction of the latent regression function f in the empirical L_2 metric. Hence, once the sieve entropy, prior mass, and prior concentration conditions are verified for the TV prior, the desired posterior contraction rate follows immediately.

12 Construction of Θ_n

Assumption. We assume that the parameter space is restricted to

$$\Theta \;=\; \Big\{\theta \in \mathbb{R}^n : \max_{1\le i\le n} |\theta_i| \le E\Big\},$$

for some fixed constant $E > 0$ chosen large enough so that $\theta_0 \in \Theta$. Thus, all candidate vectors θ are uniformly bounded in the sup-norm by E.

Let $D^{(k+1)} \in \mathbb{R}^{m\times n}$ denote the $(k+1)$-st order difference matrix, with $m = n-k-1$, and consider the prior

$$(D^{(k+1)}\theta)_\ell \overset{\text{iid}}{\sim} \text{Laplace}(0, 1/\lambda'), \qquad \ell = 1,\dots,m,$$

with $\lambda' = \lambda\frac{n^k}{k!}$ for some $\lambda > 0$. We write $\Pi_{\lambda'}$ for this prior on θ (restricted to Θ).

For a given sequence (ε_n) and a sequence (T_n) to be specified below, define the sieve

$$\Theta_n \;=\; \Big\{\theta \in \Theta : \|D^{(k+1)}\theta\|_1 \le T_n\Big\}, \qquad \Theta_n^c = \Theta \setminus \Theta_n.$$

Lemma (negligible prior mass of the sieve complement). Suppose $m = n-k-1 \asymp n$ and define

$$T_n \;\asymp\; \frac{\Big(m\log 2 + 2n\varepsilon_n^2\Big)}{n^k}$$

Then, for all n large enough,

$$\Pi_{\lambda'}(\Theta_n^c) = \Pi_{\lambda'}\big(\|D^{(k+1)}\theta\|_1 > T_n\big) \;\le\; \exp(-n\varepsilon_n^2).$$

Proof. Let

$$Z_\ell = (D^{(k+1)}\theta)_\ell, \qquad \ell = 1,\dots,m,$$

so that $Z_\ell \overset{\text{iid}}{\sim} \text{Laplace}(0, 1/\lambda)$ under Π_λ. Define

$$S_m = \sum_{\ell=1}^m |Z_\ell| = \big\|D^{(k+1)}\theta\big\|_1.$$

Then

$$\Theta_n^c = \big\{\theta \in \Theta : S_m > T_n\big\}, \qquad \Pi_\lambda(\Theta_n^c) = \mathbb{P}_\lambda(S_m > T_n).$$

We first compute an exponential moment of $|Z_\ell|$. Since Z_ℓ has density

$$f_Z(z) = \frac{\lambda'}{2}e^{-\lambda'|z|}, \qquad z \in \mathbb{R},$$

it follows that $|Z_\ell|$ is exponential with rate λ', for $k = 3$ and for any $t \in (0,\lambda)$,

$$\begin{aligned}\mathbb{E}'_\lambda\big[e^{t|Z_\ell|}\big] &= \lambda'\int_0^\infty e^{tx}e^{-\lambda' x}\,dx \\ &= \lambda'\int_0^\infty e^{-(\lambda'-t)x}\,dx = \frac{\lambda'}{\lambda'-t} < \infty.\end{aligned}$$

By independence of $\{Z_\ell\}_{\ell=1}^m$ and Markov's inequality (Chernoff bound), for any $t \in (0,\lambda')$ we have

$$\begin{aligned}\mathbb{P}'_\lambda(S_m > T_n) &= \mathbb{P}'_\lambda\Big(e^{tS_m} > e^{tT_n}\Big) \;\le\; e^{-tT_n}\,\mathbb{E}'_\lambda\big[e^{tS_m}\big] \\ &= e^{-tT_n}\Big(\mathbb{E}'_\lambda[e^{t|Z_1|}]\Big)^m \\ &= \exp\Big(-tT_n + m\log\tfrac{\lambda'}{\lambda'-t}\Big).\end{aligned}$$

Choose $t = \lambda'/2 \in (0, \lambda')$. Then

$$\frac{\lambda'}{\lambda' - t} = \frac{\lambda'}{\lambda' - \lambda'/2} = 2,$$

and hence

$$\mathbb{P}'_\lambda(S_m > T_n) \;\le\; \exp\Big(-\tfrac{\lambda'}{2} T_n + m \log 2 \Big).$$

Now take

$$T_n = \frac{2}{\lambda'} \Big(m \log 2 + 2n\varepsilon_n^2 \Big),$$

so that

$$-\tfrac{\lambda'}{2} T_n + m \log 2 = -\Big(m \log 2 + 2n\varepsilon_n^2 \Big) + m \log 2 = -2n\varepsilon_n^2.$$

Therefore,

$$\mathbb{P}_{\lambda'}(S_m > T_n) \;\le\; \exp(-2n\varepsilon_n^2) \;\le\; \exp(-n\varepsilon_n^2),$$

for all n large enough. Since $S_m = \|D^{(k+1)}\theta\|_1$ and $\Theta_n^c = \{\theta \in \Theta : S_m > T_n\}$, we conclude that

$$\Pi_{\lambda'}(\Theta_n^c) = \Pi_{\lambda'}\big(\|D^{(k+1)}\theta\|_1 > T_n \big) \;\le\; \exp(-n\varepsilon_n^2).$$

This proves that the prior mass of the sieve complement is exponentially negligible. □

13 Local KL and Variance Bounds

Lemma 1 (Upper bounds for the ALD model)**.** *Let*

$$p_{\theta_i}(y_i) = \frac{\tau(1-\tau)}{\sigma} \exp\left(-\rho_\tau\left(\frac{y_i - \theta_i}{\sigma} \right) \right), \qquad 0 < \tau < 1,\ \sigma > 0,$$

be the asymmetric Laplace density for observation Y_i, and let

$$P_\theta^n = \prod_{i=1}^n p_{\theta_i}$$

denote the joint law under the parameter vector $\theta = (\theta_1, \ldots, \theta_n)$. Let the true parameter vector be $\theta_0 = (\theta_{0,1}, \ldots, \theta_{0,n})$. Define the empirical L_2 distance

$$d_n(\theta, \theta_0) = \left(\frac{1}{n} \sum_{i=1}^n (\theta_i - \theta_{0,i})^2 \right)^{1/2}.$$

Then there exist constants $C > 0$ and $\delta_0 > 0$, depending only on τ and σ, such that

$$\max\left\{ \frac{1}{n} K(P_{\theta_0}^n, P_\theta^n),\ \frac{1}{n} V(P_{\theta_0}^n, P_\theta^n) \right\} \;\le\; C\, d_n^2(\theta, \theta_0)$$

for all θ satisfying $\|\theta - \theta_0\|_\infty \le \delta_0$.

Remark. *By standard asymptotic theory for location families, the Fisher information for the ALD provides a local quadratic approximation to the KL divergence in a neighborhood of θ_0. For sufficiently large n, the posterior concentrates on sets where this approximation holds, ensuring these bounds apply on the high posterior probability region required for the contraction argument.*

Proof. We establish the upper bounds using the Fisher information approach and properties of the asymmetric Laplace distribution.f

The log-likelihood for a single observation is

$$\ell_\theta(y) = \log p_\theta(y) = \log\left(\frac{\tau(1-\tau)}{\sigma} \right) - \frac{\rho_\tau(y - \theta)}{\sigma}.$$

The score function is

$$\frac{\partial \ell_\theta(y)}{\partial \theta} = \frac{1}{\sigma} \cdot \text{sign}_\tau(y - \theta),$$

where

$$\text{sign}_\tau(u) = \begin{cases} \tau & \text{if } u > 0, \\ \tau - 1 & \text{if } u < 0. \end{cases}$$

The Fisher information is

$$I_{\text{ALD}}(\theta) = \mathbb{E}_\theta \left[\left(\frac{\partial \ell_\theta(Y)}{\partial \theta} \right)^2 \right] = \frac{1}{\sigma^2} \mathbb{E}_\theta[\text{sign}_\tau(Y - \theta)^2].$$

For $Y \sim \text{ALD}(\theta, \sigma, \tau)$, we have $\mathbb{P}(Y < \theta) = \tau$ and $\mathbb{P}(Y > \theta) = 1 - \tau$ by construction. Therefore:

$$I_{\text{ALD}}(\theta) = \frac{1}{\sigma^2} \left[\tau^2(1 - \tau) + (\tau - 1)^2 \tau \right] = \frac{\tau(1 - \tau)}{\sigma^2} [\tau + (1 - \tau)] = \frac{\tau(1 - \tau)}{\sigma^2}.$$

By standard asymptotic theory (van der Vaart, *Asymptotic Statistics*, Theorem 7.2), for location families with finite Fisher information, the KL divergence admits the quadratic approximation

$$K(p_{\theta_0}, p_\theta) = \frac{I_{\text{ALD}}(\theta_0)}{2} (\theta - \theta_0)^2 + o((\theta - \theta_0)^2)$$

as $\theta \to \theta_0$.

More precisely, there exist constants $C_1, C_2 > 0$ and $\delta_0 > 0$ (depending only on τ and σ) such that for all $|\theta - \theta_0| < \delta_0$:

$$K(p_{\theta_0}, p_\theta) \leq \frac{2\tau(1 - \tau)}{\sigma^2} (\theta - \theta_0)^2.$$

Since observations are independent, the joint KL divergence is

$$K(P_{\theta_0}^n, P_\theta^n) = \sum_{i=1}^n K(p_{\theta_{0,i}}, p_{\theta_i}).$$

Applying the single-observation bound to each term, whenever $\|\theta - \theta_0\|_\infty < \delta_0$:

$$K(P_{\theta_0}^n, P_\theta^n) \leq \frac{2\tau(1 - \tau)}{\sigma^2} \sum_{i=1}^n (\theta_i - \theta_{0,i})^2 = \frac{2\tau(1 - \tau)}{\sigma^2} \cdot n \cdot d_n^2(\theta, \theta_0).$$

Dividing by n:

$$\frac{1}{n} K(P_{\theta_0}^n, P_\theta^n) \leq \frac{2\tau(1 - \tau)}{\sigma^2} d_n^2(\theta, \theta_0).$$

By Le Cam's inequality (Tsybakov, *Introduction to Nonparametric Estimation*, Lemma 2.5), for any two probability measures P and Q:

$$V(P, Q) = \text{Var}_P \left(\log \frac{dP}{dQ} \right) \leq 2K(P, Q).$$

Applying this to each observation:

$$V(p_{\theta_{0,i}}, p_{\theta_i}) \leq 2K(p_{\theta_{0,i}}, p_{\theta_i}).$$

For the product measure, since observations are independent:

$$V(P_{\theta_0}^n, P_\theta^n) = \sum_{i=1}^n V(p_{\theta_{0,i}}, p_{\theta_i}) \leq 2 \sum_{i=1}^n K(p_{\theta_{0,i}}, p_{\theta_i}) = 2K(P_{\theta_0}^n, P_\theta^n).$$

Combining with the KL bound:

$$\frac{1}{n} V(P_{\theta_0}^n, P_\theta^n) \leq \frac{2}{n} K(P_{\theta_0}^n, P_\theta^n) \leq \frac{4\tau(1 - \tau)}{\sigma^2} d_n^2(\theta, \theta_0).$$

Taking $C = \max\left\{\frac{2\tau(1-\tau)}{\sigma^2}, \frac{4\tau(1-\tau)}{\sigma^2}\right\} = \frac{4\tau(1-\tau)}{\sigma^2}$, we obtain

$$\max\left\{\frac{1}{n}K(P_{\theta_0}^n, P_\theta^n),\ \frac{1}{n}V(P_{\theta_0}^n, P_\theta^n)\right\} \ \le\ C\, d_n^2(\theta, \theta_0)$$

for all θ satisfying $\|\theta - \theta_0\|_\infty \le \delta_0$, where δ_0 is the radius of validity for the Fisher information approximation.

Note that $C > 0$ for all $\tau \in (0,1)$, since $\tau(1-\tau) > 0$ on this interval. □

14 Prior Mass Lower Bound for TV Prior

Theorem 3 (Prior Concentration Lower Bound)**.** *Let $\theta \in \mathbb{R}^n$ with prior $\pi(D^k\theta) \propto \exp(-\frac{n^k}{k!}\|D^k\theta\|_1)$, where $(D^k\theta)_i \overset{iid}{\sim}$ Laplace$(0, 1/\lambda')$.*

Let θ_0 be the true parameter with $s_n = \|D^k\theta_0\|_0$ nonzero entries. Then

$$\pi(\|\theta - \theta_0\|_2 < \epsilon) \gtrsim \exp\left(-s_n \log n - s_n \log(1/\epsilon) - \lambda s_n \|D^k\theta_0\|_\infty\right).$$

Proof. The third-order difference operator $D^k : \mathbb{R}^n \to \mathbb{R}^{n-k}$ has pseudoinverse $(D^k)^\dagger$ satisfying

$$\|\theta - \theta_0\|_2 \le \|(D^k)^\dagger\|_{\text{op}} \cdot \|D^k(\theta - \theta_0)\|_2.$$

For the discrete k-th order difference operator, the operator norm of the pseudoinverse satisfies $\|(D^k)^\dagger\|_{\text{op}} = O(n^k)$.

Therefore, there exists a constant $C > 0$ such that

$$\|\theta - \theta_0\|_2 \le Cn^k \|D^k\theta - D^k\theta_0\|_2.$$

To ensure $\|\theta - \theta_0\|_2 < \epsilon$, it suffices to have

$$\|D^k\theta - D^k\theta_0\|_2 < \delta \quad \text{where} \quad \delta := \frac{\epsilon}{Cn^k}.$$

Let $S_n = \{i : (D^k\theta_0)_i \neq 0\}$ denote the support of $D^k\theta_0$, with $|S_n| = s_n$. Define the slab

$$\mathcal{S} = \left\{D^k\theta \in \mathbb{R}^{n-k} : |(D^k\theta)_i - (D^k\theta_0)_i| < r \text{ for all } i \in S_n\right\}$$

where $r > 0$ is to be chosen.

For any $D^k\theta \in \mathcal{S}$:

$$\begin{aligned}
\|D^k\theta - D^k\theta_0\|_2^2 &= \sum_{i=1}^{n-k} |(D^k\theta)_i - (D^k\theta_0)_i|^2 \\
&= \sum_{i \in S_n} |(D^k\theta)_i - (D^k\theta_0)_i|^2 + \sum_{i \notin S_n} |(D^k\theta)_i - 0|^2 \\
&< s_n r^2 + \sum_{i \notin S_n} |(D^k\theta)_i|^2.
\end{aligned}$$

Set $r = \delta/\sqrt{s_n}$. Then the signal coordinates contribute

$$\sum_{i \in S_n} |(D^k\theta)_i - (D^k\theta_0)_i|^2 < s_n r^2 = s_n \cdot \frac{\delta^2}{s_n} = \delta^2.$$

Therefore

$$\Pi(\|D^k\theta - D^k\theta_0\|_2 \ < 2\delta) = \Pi\left(\sum_{i \in S_n} |(D^k\theta)_i - (D^k\theta_0)_i|^2 < \delta^2\right) \cdot \Pi\left(\sum_{i \notin S_n} |(D^k\theta)_i|^2 < \delta^2\right)$$

To compute the first component, notice that since the $(D^k\theta)_i$ are independent across i then:

$$\pi(\mathcal{S}) = \prod_{i \in S_n} \mathbb{P}\left(|(D^k\theta)_i - (D^k\theta_0)_i| < r\right)$$

For each $i \in S_n$, with $(D^k\theta)_i \sim \text{Laplace}(0, 1/\lambda')$ having density $f(u) = \frac{\lambda'}{2}e^{-\lambda'|u|}$:

$$\mathbb{P}\left(|(D^k\theta)_i - (D^k\theta_0)_i| < r\right) = \int_{(D^k\theta_0)_i - r}^{(D^k\theta_0)_i + r} \frac{\lambda'}{2} e^{-\lambda'|u|}\, du.$$

Lower bound for the integral: Assume without loss of generality that $(D^k\theta_0)_i > 0$ and $(D^k\theta_0)_i > r$ (signal is sufficiently large relative to the radius). Then for $u \in [(D^k\theta_0)_i - r, (D^k\theta_0)_i + r]$, we have $u > 0$ and

$$e^{-\lambda'|u|} = e^{-\lambda' u} \geq e^{-\lambda'((D^k\theta_0)_i + r)}.$$

Therefore:

$$\begin{aligned}
\mathbb{P}\left(|(D^k\theta)_i - (D^k\theta_0)_i| < r\right) &\geq \frac{\lambda'}{2} \int_{(D^k\theta_0)_i - r}^{(D^k\theta_0)_i + r} e^{-\lambda'((D^k\theta_0)_i + r)}\, du \\
&= \frac{\lambda'}{2} \cdot 2r \cdot e^{-\lambda'((D^k\theta_0)_i + r)} \\
&= \lambda' r \cdot e^{-\lambda'(D^k\theta_0)_i} \cdot e^{-\lambda' r}.
\end{aligned}$$

For small $\lambda' r$ (specifically, when $\lambda' r \ll 1$), we have $e^{-\lambda' r} \geq c$ for some absolute constant $c > 0$. Thus:

$$\mathbb{P}\left(|(D^k\theta)_i - (D^k\theta_0)_i| < r\right) \geq c\lambda' r \cdot e^{-\lambda'|(D^k\theta_0)_i|}.$$

$$\begin{aligned}
\pi(\mathcal{S}) &\geq \prod_{i \in S_n} c\lambda' r \cdot e^{-\lambda'|(D^k\theta_0)_i|} \\
&= (c\lambda' r)^{s_n} \cdot \prod_{i \in S_n} e^{-\lambda'|(D^k\theta_0)_i|} \\
&= (c\lambda' r)^{s_n} \cdot \exp\left(-\lambda' \sum_{i \in S_n} |(D^k\theta_0)_i|\right) \\
&= (c\lambda' r)^{s_n} \cdot \exp(-\lambda' \|D^k\theta_0\|_1).
\end{aligned}$$

Using $\|D^k\theta_0\|_1 = \sum_{i \in S_n} |(D^k\theta_0)_i| \leq s_n \|D^k\theta_0\|_\infty$:

$$\pi(\mathcal{S}) \geq (c\lambda' r)^{s_n} \cdot \exp(-\lambda' s_n \|D^k\theta_0\|_\infty).$$

Recall that $r = \delta/\sqrt{s_n} = \epsilon/(Cn^k\sqrt{s_n})$. Substituting:

$$\begin{aligned}
\log \pi(\mathcal{S}) &\geq s_n \log(c\lambda' r) - \lambda' s_n \|D^k\theta_0\|_\infty \\
&= s_n \log\left(\frac{c\lambda'\epsilon}{Cn^k\sqrt{s_n}}\right) - \lambda' s_n \|D^k\theta_0\|_\infty \\
&= s_n \log(c\lambda'\epsilon) - s_n \log C - s_n \log(n^k) - s_n \log(\sqrt{s_n}) - \lambda' s_n \|D^k\theta_0\|_\infty \\
&= s_n \log(c\lambda'\epsilon) - s_n \log C - k s_n \log n - \frac{s_n}{2} \log s_n - \lambda' s_n \|D^k\theta_0\|_\infty.
\end{aligned}$$

The terms $s_n \log(c\lambda'\epsilon)$ and $-s_n \log C$ absorb into constants. When $s_n = o(n)$, the term $\frac{s_n}{2} \log s_n$ is $o(s_n \log n)$ and can be absorbed. Therefore:

$$\log \pi(\|\theta - \theta_0\|_2 < \epsilon) \gtrsim -k s_n \log n - s_n \log(1/\epsilon) - \lambda' s_n \|D^k\theta_0\|_\infty.$$

Up to constants in the $\log n$ coefficient:

$$\pi(\|\theta - \theta_0\|_2 < \epsilon) \gtrsim \exp\left(-s_n \log n - s_n \log(1/\epsilon) - \lambda' s_n \|D^3\theta_0\|_\infty\right).$$

□

We now determine under what conditions the prior mass is lower bounded by $\exp(-Cn\epsilon_n^2)$, which is the typical requirement for posterior contraction results.

Proposition 1 (Condition for the Rate)**.** *The prior satisfies*

$$\pi(\|\theta - \theta_0\|_2 < \epsilon_n) \gtrsim \exp(-Cn\epsilon_n^2)$$

if and only if

$$s_n \log n + s_n \log(1/\epsilon_n) + \frac{n^k}{k!} s_n \|D^3\theta_0\|_\infty \lesssim n\epsilon_n^2. \tag{95}$$

Proof. From Theorem 1, we have

$$\log \pi(\|\theta - \theta_0\|_2 < \epsilon_n) \gtrsim -s_n \log n - s_n \log(1/\epsilon_n) - \lambda s_n \|D^3\theta_0\|_\infty.$$

For the lower bound $\pi(\|\theta - \theta_0\|_2 < \epsilon_n) \gtrsim \exp(-Cn\epsilon_n^2)$ to hold, we need

$$-s_n \log n - s_n \log(1/\epsilon_n) - \lambda s_n \|D^3\theta_0\|_\infty \gtrsim -Cn\epsilon_n^2,$$

which is equivalent to

$$s_n \log n + s_n \log(1/\epsilon_n) + \lambda s_n \|D^3\theta_0\|_\infty \lesssim Cn\epsilon_n^2.$$

□

Under standard regularity assumptions where $\|D^3\theta_0\|_\infty = O(1)$ and $\lambda = O(1)$, condition (95) simplifies to:

$$s_n \left[\log n + \log(1/\epsilon_n)\right] \lesssim n\epsilon_n^2.$$

Ignoring logarithmic factors, the condition becomes:

$$\boxed{s_n \lesssim n\epsilon_n^2.}$$

Interpretation: The sparsity level s_n (number of knots) must not grow faster than $n\epsilon_n^2$ for the prior to achieve contraction rate ϵ_n.

15 References to follow up on

- [3]
- [4]
- [1]
- [2]
- [5]

References

[1] Richard H. Gerlach, Cathy W. S. Chen, and Nancy Y. C. Chan. Bayesian time-varying quantile forecasting for value-at-risk in financial markets. *Journal of Business & Economic Statistics*, 29(4):481–492, 2011.

[2] Cai Li and Heping Zhang. Tensor quantile regression with application to association between neuroimages and human intelligence. *The Annals of Applied Statistics*, 15(3):1455 – 1477, 2021.

[3] Takahiro Onizuka, Shintaro Hashimoto, and Shonosuke Sugasawa. Fast and locally adaptive bayesian quantile smoothing using calibrated variational approximations. *Statistics and Computing*, 34(1):15, 2023.

[4] Takahiro Onizuka, Shintaro Hashimoto, and Shonosuke Sugasawa. Locally adaptive spatial quantile smoothing: Application to monitoring crime density in tokyo. *Spatial Statistics*, 59:100793, 2024.

[5] Oscar Hernan Madrid Padilla and Sabyasachi Chatterjee. Quantile regression by dyadic CART. *Electronic Journal of Statistics*, 18(1):1206 – 1247, 2024.